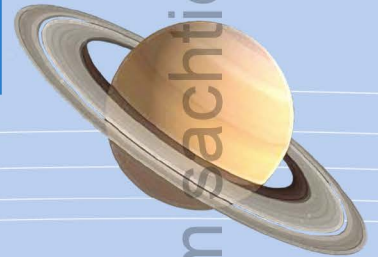
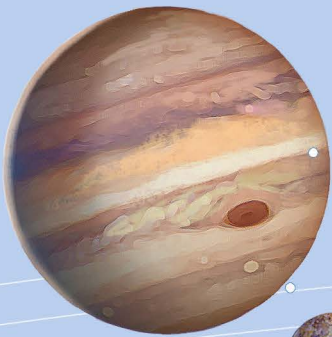


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CAMBRIDGE Primary Science

Workbook 6

Fiona Baxter & Liz Dilley



Second edition

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Workbook 6

Fiona Baxter & Liz Dilley

Imprint

TO FOLLOW ...

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How to use this book



This workbook provides questions for you to practise what you have learned in class. There is a topic to match each topic in your Learner's Book. Each topic contains the following sections:

Focus: these questions help you to master the basics



Focus

- 1 Explain why the pencil in the picture appears to be bent. Cross out the incorrect alternatives in the sentences below.

The pencil is bent because of reflection/refraction.

A ray of light passes from the pencil through the water/air to the glass. The ray bends/straightens when it passes through the glass to the air/water and into our eyes.

We see the bent pencil as a trick/optical illusion.



Practice: these questions help you to become more confident in using what you have learned



Practice

Circle the letter of the correct answer to each of the following questions.

- 3 Your heart pumps blood through the body. This process is called ...
 - A heartbeat
 - B circulation
 - C pulsing
- 4 The circulatory system is made up of the ...
 - A heart only
 - B heart and blood vessels
 - C heart, blood vessels and blood

Challenge: these questions will make you think more deeply



Challenge

Class 5 measured the pulse rate and breathing rate of 10 people after they had jogged on the spot for three minutes. Here are their results.

Person	Breathing rate (breaths per minute)	Pulse rate (heartbeats per minute)
1	30	90
2	50	120
3	35	102
4	32	95
5	26	100
6	40	110
7	45	115
8	33	98
9	38	106
10	42	112

1

The human body

> 1.1 The circulatory system

Focus

1 Name the three parts of the circulatory system.

a _____

b _____

c _____

2 Use the words in the box to complete the sentences.
You will use some words more than once.

blood vessels

oxygen

blood

waste products

food

lungs

a The heart pumps _____ through the body.

b The left side of the heart pumps _____ that
contains _____.

c The right side of the heart pumps _____
without _____ to the _____.

d Blood is carried in the _____.

e Blood carries _____ and _____ to
all parts of the body and takes away _____.

Practice

- 3 Circle the letter of the correct answer to each of the following questions.
- a Your heart pumps blood through the body.
This process is called ...
 - A heartbeat
 - B circulation
 - C pulsing
 - b The circulatory system is made up of the ...
 - A heart only
 - B heart and blood vessels
 - C heart, blood vessels and blood
 - c The left side of the heart pumps blood that contains ...
 - A oxygen
 - B no oxygen
 - C many different gases
 - d The right side of the heart pumps blood to ...
 - A the brain
 - B the lungs
 - C the kidneys
 - e Which blood vessels bring oxygen to all the body cells and carry away wastes?
 - A arteries
 - B veins
 - C capillaries

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Challenge

- 4 Marcus measured his pulse rate while he was sitting still and then after doing different types of physical activities. These are his results.

Activities	Heartbeats per minute
Sitting still	72
Running on the spot	120
Playing football	150
Digging in the garden	80
Riding a bicycle	110

- a Draw a bar chart of Marcus's results.
Use a different colour pencil for each activity.



- b When was Marcus's pulse rate lowest? Explain why.

- c Which activity caused the highest pulse rate?

- d Is there a measurement that needs to be checked?
If so, which one and why?

- e Explain how Marcus could make this a fair test.

- f Write a conclusion for Marcus's findings.

- g Predict how Marcus's pulse rate would change if he pedalled faster on the bicycle. Explain your answer.

> 1.2 The respiratory system

Focus

- 1 Use the words in the box to complete the sentences.
You will use some words more than once.

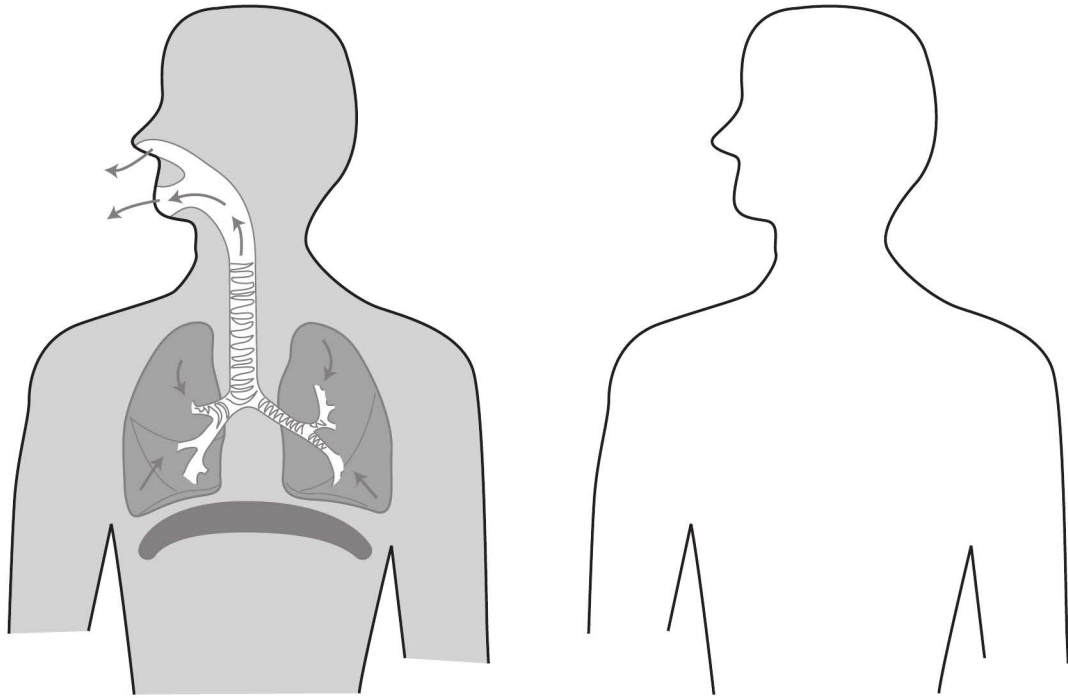
blood	ribs	lungs	windpipe
nose	carbon dioxide	oxygen	

We breathe in air through our _____. The air we breathe in contains _____ gas. The air moves down the _____ and into our _____.

The _____ in the air then moves from the _____ into the _____. We breathe out air that contains _____ gas. The _____ protect our respiratory system.

Practice

- 2 The drawing shows the lungs when you breathe out. Make a drawing to show what happens to the lungs when you breathe in. Add labels to your drawing to explain what it shows.



- 3 Complete the flow diagram using these words to show the path of oxygen when we breathe in.

lungs nose blood windpipe

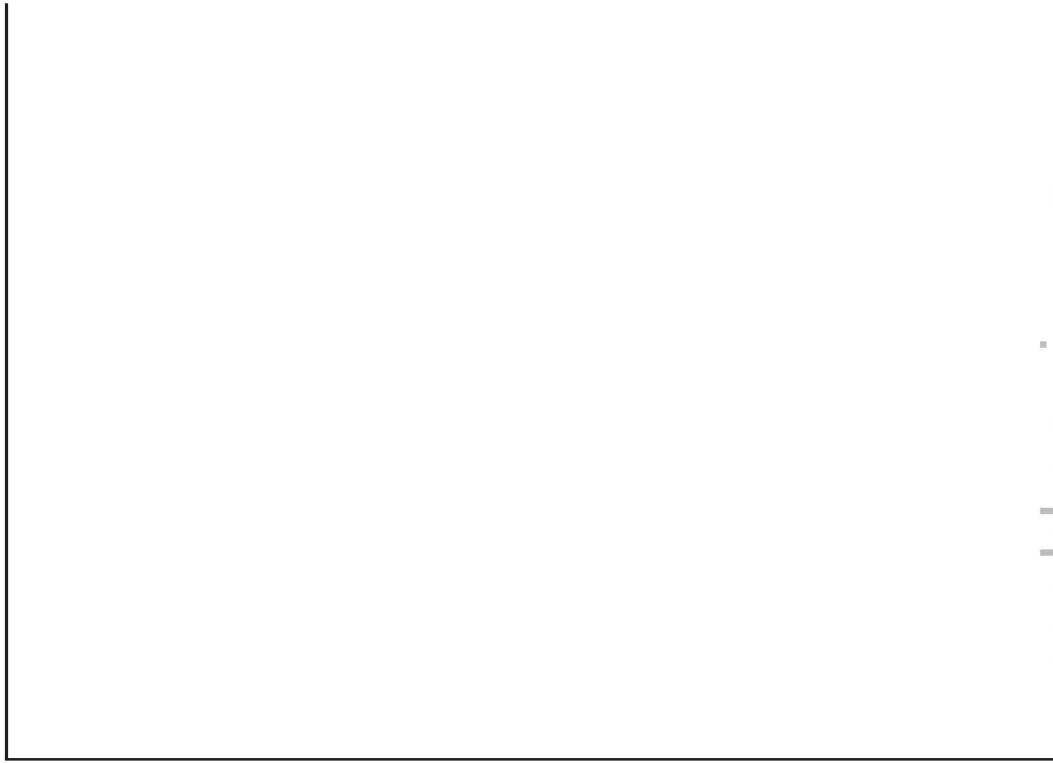


Challenge

- 4 Class 5 measured the pulse rate and breathing rate of 10 people after they had jogged on the spot for three minutes. Here are their results.

Person	Breathing rate (breaths per minute)	Pulse rate / heartbeats per minute
1	30	90
2	50	120
3	35	102
4	32	95
5	26	100
6	40	110
7	45	115
8	33	98
9	38	106
10	42	112

- a Draw a scatter graph of the results.



- b Describe the pattern you observe in the results.

- c i Identify any results that do not fit the pattern.

- ii Suggest a reason for this.

- d Use the graph to predict the pulse rate of a person whose breathing rate is 48 heartbeats per minute

- e Suggest a conclusion that class 5 can make from these results.

> 1.3 The reproductive system

Focus

- 1 Match each of the words in List 1 with their meanings in List 2.
Draw a line to link each word to its meaning.

List 1	List 2
reproduction	the baby develops here
puberty	male sex cells are made here
ovum	female sex cells are made here
fertilisation	male sex cell
uterus	female sex cell
testis	making more individuals of the same kind of living thing
sperm	joining of a male sex cell and female sex cell
ovary	the age at which a person becomes able to reproduce

Practice

2 Draw a circle around the letter of the correct answer to each of these questions.

a Which change in puberty happens to both boys and girls?

A *shoulders and chest get broader*

B *hips get wider*

C *the body grows more hair*

b Which change in puberty happens to boys only?

A *increase in height*

B *voice gets much deeper*

C *skin becomes more oily*

c Which one of the following is not part of the female reproductive system?

A *ovary*

B *testes*

C *uterus*

d Which one of the following is not part of the male reproductive system?

A *oviduct*

B *sperm duct*

C *penis*

e Which of the following happens during menstruation?

A *male and female sex cells join*

B *new eggs are formed*

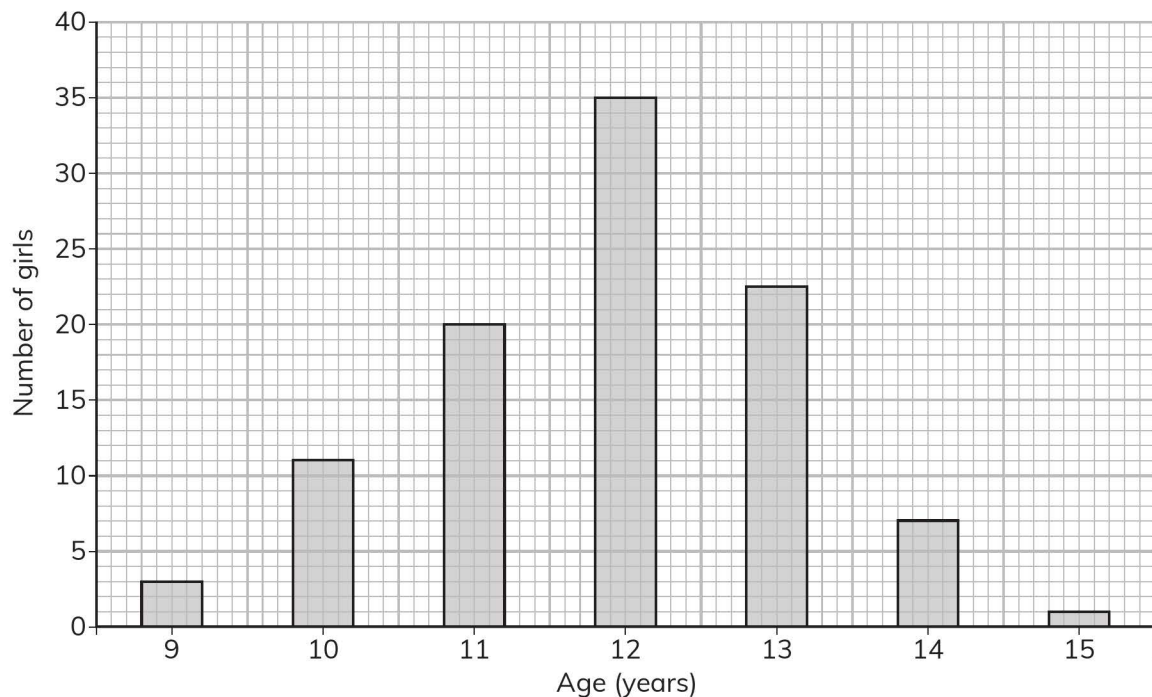
C *the lining of the uterus pulls away*

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Challenge

- 3 A scientist carried out a study with 100 girls to find out the age at which they started puberty.

This a graph of her findings.



- a Describe three signs in girls that puberty has started.

- b i At what age did most of the 100 girls start puberty?

- ii How many girls started puberty at this age?

c How many girls started puberty at the age of:

i 10?

ii 14?

d The scientist also measured the mass of the girls at the start of puberty. This is a summary of her results.

age	average mass / kg
9	45
10	45
11	46
12	46
13	47
14	46
15	45

i What pattern can you see in the results?

ii Suggest a factor that affects the age at which puberty starts.

- iii In the past, most girls started puberty at about the age of 14. Suggest a reason why puberty starts at a younger age.

> 1.4 Diseases

Focus

- 1 Mark each one of these statements about diseases as true ✓ or false ✗.

a All infectious diseases are caused by viruses.

☐

b A parasite lives on or in the body of another living thing.

☐

c Living things that spread diseases always get the disease themselves.

☐

d Diseases can be spread when we cough or sneeze.

☐

e Washing hands with soap and water will stop germs spreading.

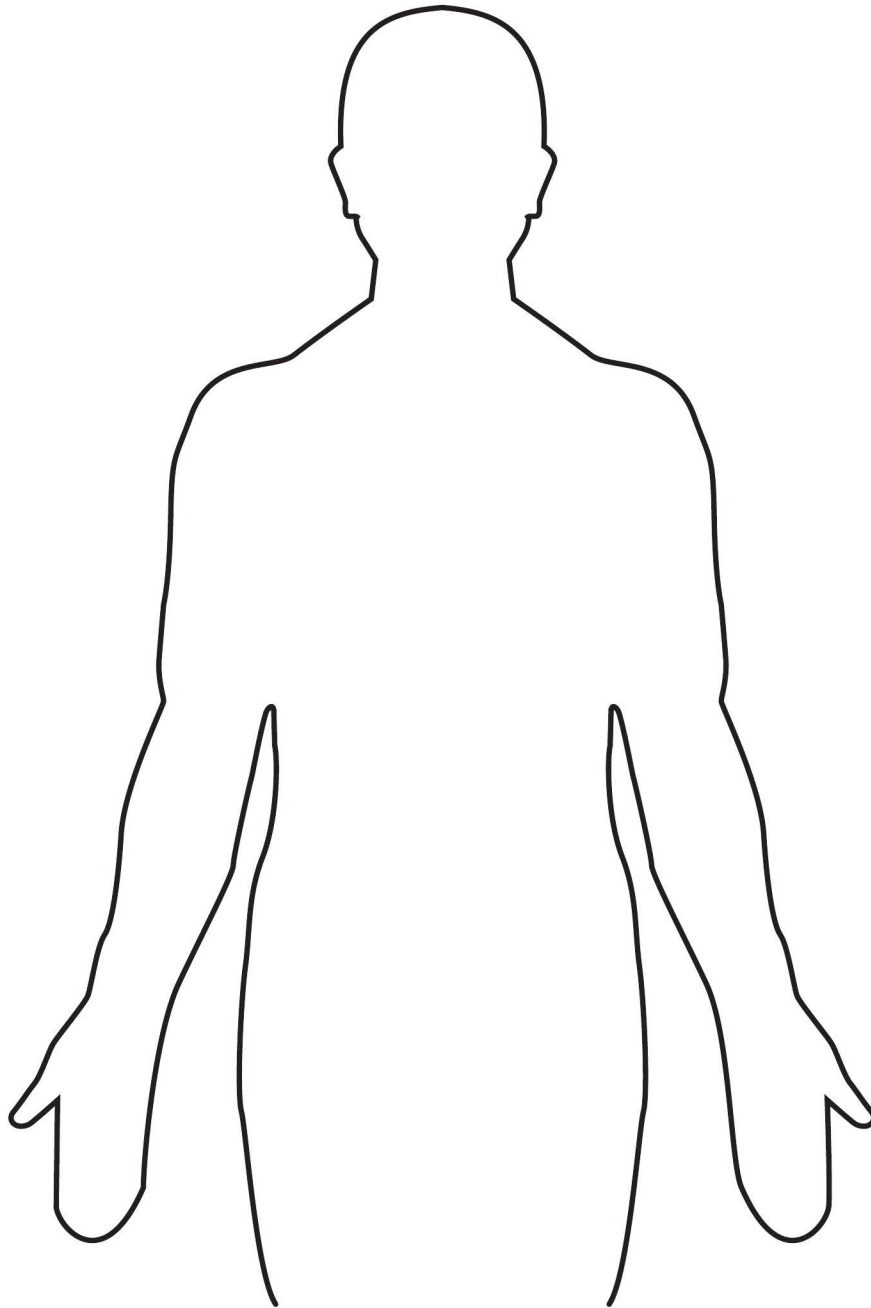
☐

f Adding salt to water will make it safe to drink.

☐

Practice

- 2 On the outline of the human body, draw in the parts and write labels to describe the different ways the body defends itself against infectious diseases.



PDF from sachtienganhanoi.com

Challenge

- 3 Malaria is a serious disease that kills millions of people around the world each year. The table shows the number of malaria cases reported at a clinic in Indonesia in one year.

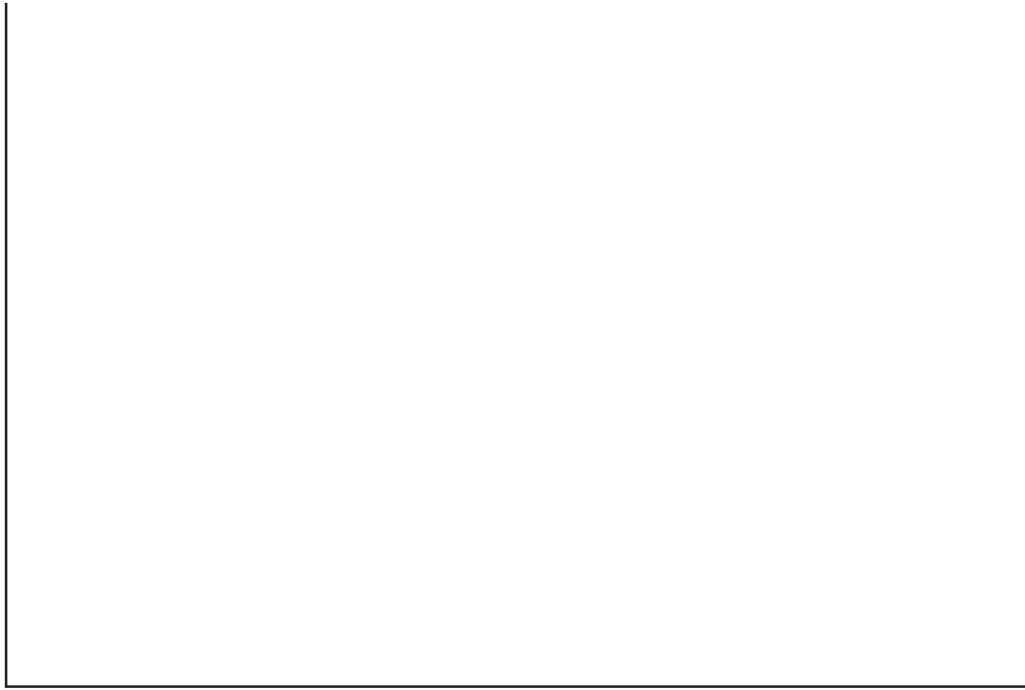
Indonesia has a warm tropical climate with a dry season from April to October, and a wet monsoon season from November to March.

Months of the year	Number of malaria cases reported
January to March	780
April to June	120
July to September	79
October to December	326

- a How is malaria caused?

- b Mosquitoes are vectors for malaria What does this mean?

- c i Draw a bar chart to show the data in the table.



- ii In which months of the year were the most malaria cases reported?

- iii In which months of the year were the fewest malaria cases reported?

- iv Suggest reason for this pattern in the data

- d Describe three things that people can do to help prevent getting diseases such as malaria.

2

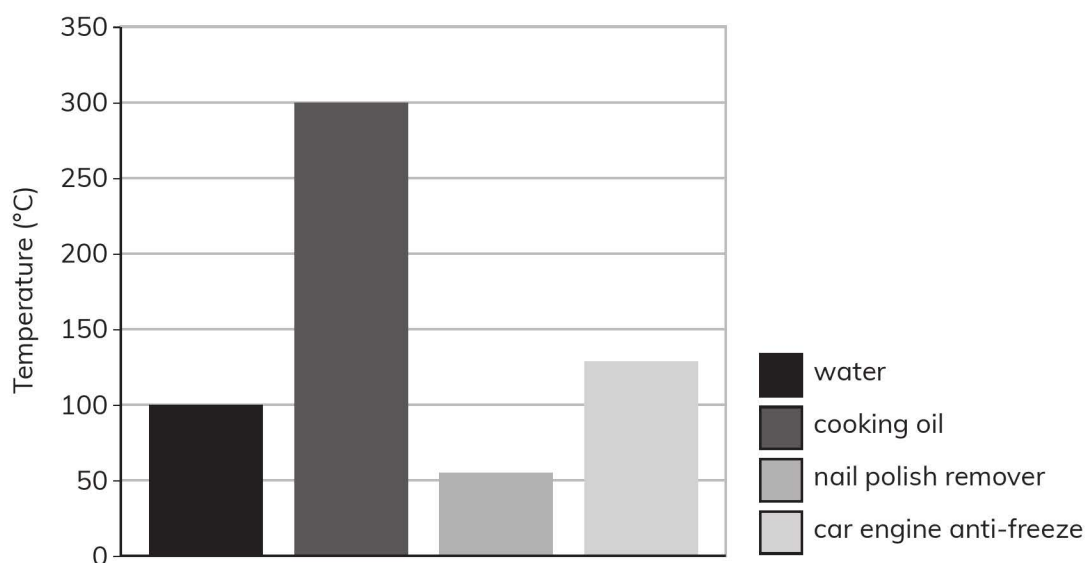
Materials: properties and changes

> 1.1 Properties of substances

Focus

- 1 Different liquids boil at different temperatures.
Explain why this is so.

- 2 The graph shows the temperatures at which some liquids boil.



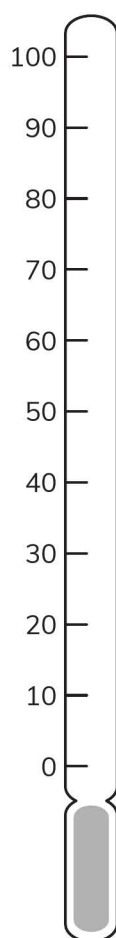
- a Which liquid boils at the highest temperature?

- b Which liquid boils at the lowest temperature?

c At what temperature does car engine anti-freeze boil?

d Name another process in which the same change of state takes place.

3 Mark off the following temperatures on the thermometer.



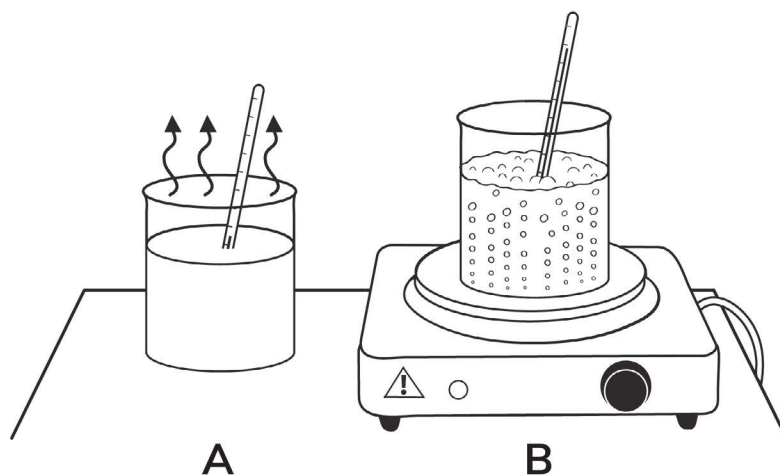
Boiling point of acetone: 50.5°C

Melting point of candle wax: 60°C

Boiling point of water: 100°C

Practice

4 Look at the pictures of beakers of water in a classroom.



a Which drawing shows boiling?

b What process does the other drawing show?

c i Write down one thing that is the same about the two processes.

ii Write down two things that are different about the two processes.

d Predict what you think the temperature will be in:

i Beaker A _____

ii Beaker B _____

e Explain the reason for your answer to 4b

Challenge

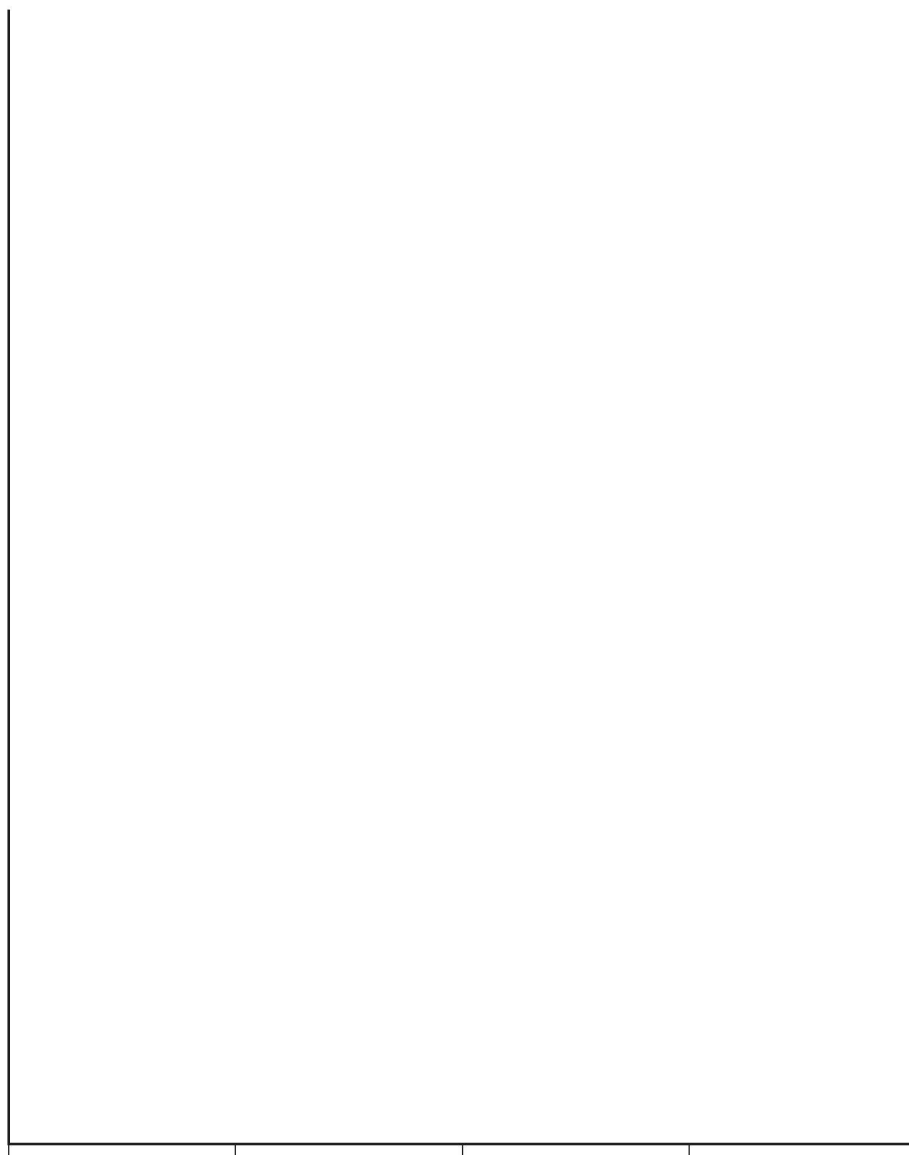
5 These are the boiling points of three liquids:

Liquid	Boiling point in °C
Water	100
Olive oil	300
Vinegar	118

a Order the liquids from the one with the highest boiling point to the one with the lowest boiling point.

b If 100 ml of all three liquids are heated on the same Bunsen burner for the same amount of time, which one will turn into a gas first? Do not heat any liquids yourself.

- c Draw a bar graph to compare the boiling points of water, vinegar and olive oil.

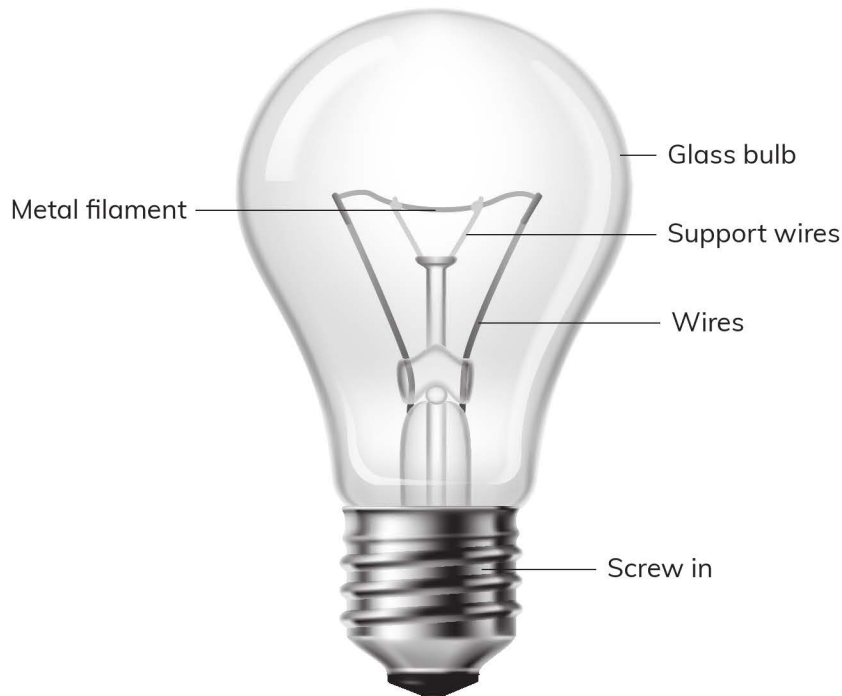


- d Find out the boiling point of iodine and add it to your graph.
- e Why is working with boiling oil more dangerous than working with boiling water?

> 2.2 Thermal and electrical conductors

Focus

1 Look at the picture of a lamp.



a i Which parts of the lamp are electrical conductors?

ii Which part of the lamp is not an electrical conductor?

b Why are electrical plugs made of plastic?

- c i Why could your hand get burnt if you stirred hot soup with a metal spoon?

- ii Why is a wooden spoon good for stirring soup cooking in a pot?

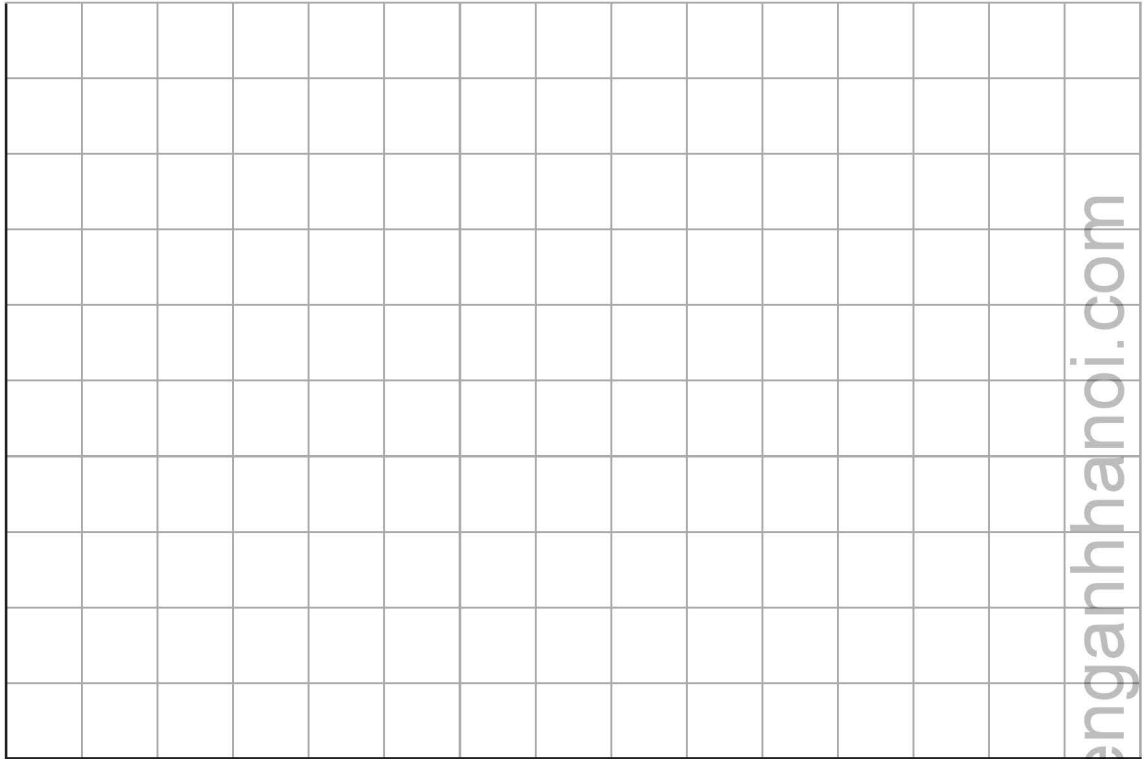
Practice

- 2 The table shows how well some common metals conduct heat. A high value means that the metal conducts heat very well.

Metal	Measure of how well the metal conducts heat
silver	420
brass	109
copper	400
iron	80
stainless steel	15
aluminium	250

- a i List the metals in order from best conductor of heat to worst conductor of heat.

ii Draw a graph of the data.



b Why do some cooking pots have a copper bottom?

c Why do you think cooking pots are not made from silver?

d Why are car radiators which cool the engine made from copper and brass?

Challenge

- 3 a Will a block of ice melt quicker in a plastic dish or a metal dish of the same size? Say why.

- b If you touched a metal dish that contained an ice cube and a plastic dish that contained an ice cube, which dish would feel warmer? Say why.

- 4 Class 6 asked the question: *Is graphite an electrical conductor or not?* They carried out an investigation to find out. Each group tested different lengths of graphite in a circuit and measured the brightness of a lamp in circuit on scale of 0–5. Averages of their results are shown in the table.

Length of graphite / cm	Average brightness
10	1
8	2
2	5
6	4
4	3
12	0

a Identify the variables in the investigation:

dependent variable

independent variable

control variables

b Draw a graph of the data



c Why did they work an average brightness for each length of graphite tested?

- d Describe the pattern seen in the results.

- e Predict the brightness of the lamp if Class 6 tested a 1 cm length of graphite. Add your predicted brightness to the graph.

- f What is the answer to class 6's question?

> 2.3 Reversible changes

Focus

- 1 Say whether each of these changes is reversible or irreversible:

- a melting butter in a hot pan

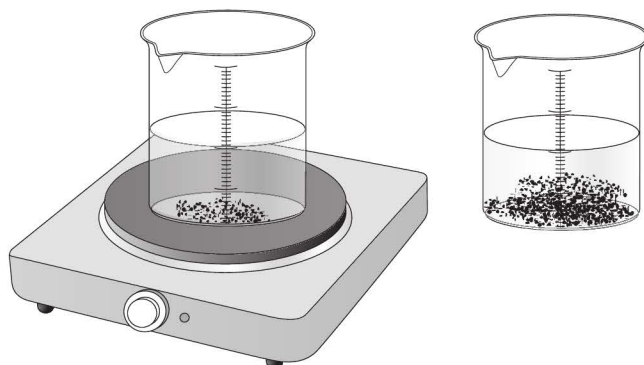
- b baking a cake

- c dissolving salt in water

- d burning wood on a fire

- e rusting on a tin can

- 2 The picture shows a test to find out about factors that affect dissolving.



- a Which factor that affects dissolving is being tested?

- b Which factors are the same in both beakers?

- c Which factors are different in the beakers?

- d Will this be a fair test? Explain your answer.

Practice

- 3 Mrs Pather poured a cup of tea from the teapot and added two teaspoons of sugar and stirred the a few times. The tea was warm, not hot, so she drank it quickly. As she drank the last few drops, she noticed there was still sugar in the bottom of the cup.



- a Why was there still sugar at the bottom of the cup?

- b i What two things could Mrs Pather have done to make the all sugar dissolve?

- ii Use your knowledge about particles to explain how these two things help sugar dissolve faster.

- iii Draw a picture to show what happens to the particles of substances when a solid dissolves in a liquid. Labels the particles.

- c i Predict what would happen to the sugar if Mrs Pather forgot to drink her cup of tea and left it standing for the whole afternoon.

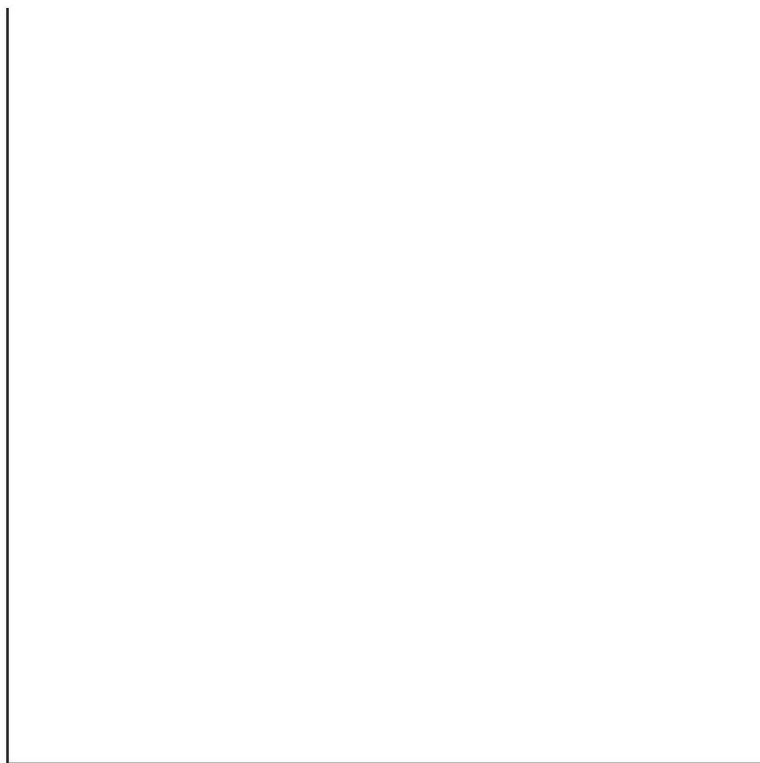
- ii Use the particle model to explain your prediction.

Challenge

- 4 Zara and Sofia carried out a fair test on the effect of temperature on the rate of dissolving. These are their results.

Water temperature / °C	Time for sugar to dissolve / seconds
20	90
40	40
60	25
80	15

- a Identify the following variables in their test:
- i the dependent variable:
 - ii the independent variable:
 - iii two controlled variables:
- b Draw a graph of the results. Remember to label the axes and give the graph a heading.



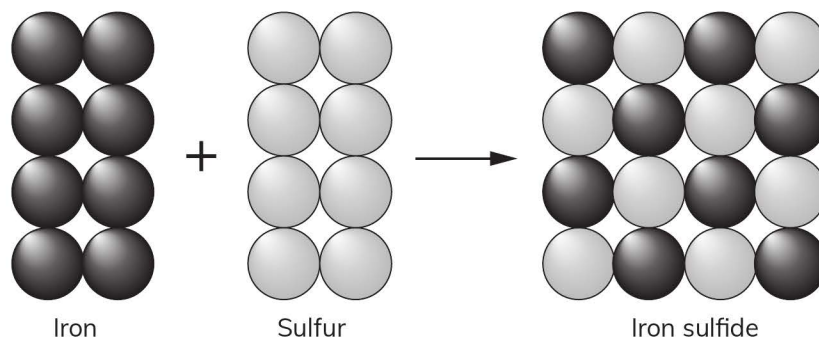
- c Describe the pattern you can see in their results.

- d What could Zara and Sofia conclude from their investigation?

> 2.4 Chemical reactions

Focus

1 The drawing shows what happens when we mix iron and sulfur.



a Say why this is a chemical reaction.

b Name the reactants in the reaction.

c Name the product that forms.

2 Describe three ways we can tell if a chemical reaction has taken place.

Practice

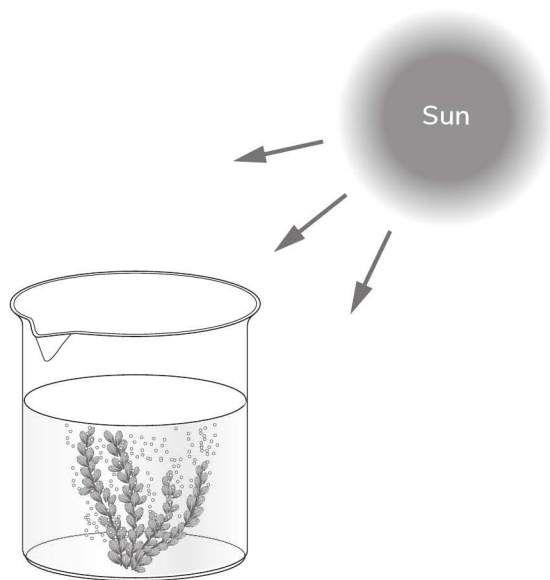
3 Circle the correct answer to each of the questions.

- a When cooking gas burns in oxygen, carbon dioxide and water are formed. Which is a reactant?
- A water
B oxygen
C heat
- b When we mix magnesium metal with oxygen, we see a bright white flame and a white powder. Which is a product?
- A the flame
B the powder
C oxygen
- c When we mix vinegar and baking soda, carbon dioxide gas and water are formed. What evidence do we have that a chemical reaction has taken place?
- A We can taste it to check for water.
B We can see a colour change.
C We can see gas bubbles form.
- d Eggs rotting is an example of a chemical reaction because ...
- A a gas that smells bad is formed
B the egg shrinks
C the egg yolk and egg white separate from each other.
- e Which of the following is not evidence that a chemical reaction has occurred?
- A formation of liquid droplets above the solution
B colour of solution changing from clear to yellow
C formation of smoke when a candle burns

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VIGNETTE

Challenge

- 4 When plants make food, they use carbon dioxide gas and water to form oxygen and a sugar called glucose. The picture shows a plant called pond weed.



- a i Identify the reactants in the reaction described.

- ii Identify the products in the reaction described.

- b i Suggest a way that we can obtain evidence that the chemical reaction has taken place.

- ii How could we use this evidence to determine the rate of the reaction?

- c Where do plants living on land get these reactants from?

- d Why is the chemical reaction described important for living things?

- e Find out:

- i the name of the process by which plants make food

- ii any other factors that plants need for this process.

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3

Rocks, the rock cycle and soil

> 3.1 Igneous rocks

Focus

- 1 Complete these sentences by writing in the correct words.

Choose words from the box:

minerals	crystalline	magma	intrusive
extrusive	basalt	granite	quickly
slowly	crystals	small	crust

Igneous rocks form when _____ or lava cool

down into solid rock. An _____ igneous

rock forms when magma cools down inside the Earth's

_____. The magma cools _____.

This causes the _____ to be large. An example

of one of these types of rock is _____.

When lava cools down on the surface of the Earth,

an _____ igneous rock forms. The lava cools

_____ and the crystals are very _____.

An example of one of these types of rock is _____.

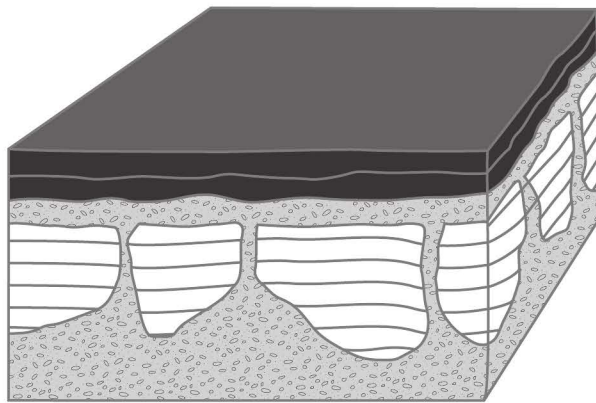
All igneous rocks are formed of crystals of _____.

These give the rock a _____ appearance.

Practice

2 The diagram shows a section of the Earth's crust and the surface of the Earth.

a Label the following features: Intrusive igneous rock; Extrusive rock



b Give an example of each type of rock you have labelled.

c Describe the difference between the two rocks you named.

d Explain this difference.

Challenge



- 3 Arun and Marcus have gone to an island on holiday. There is a big mountain in the middle of the island. They are walking along the beach. The beach is covered with black stones that have been smoothed by the sea.

The boys have been learning about igneous rocks at school.

‘I think these stones are granite,’ says Arun.

‘No, Arun, they are basalt,’ says Marcus.

- a Who is correct, Arun or Marcus?

- b Give reasons why he is correct.

- c Describe why you think this rock is on the island.
(Hint: read about the island above for a clue.)

> 3.2 Sedimentary rocks and fossils

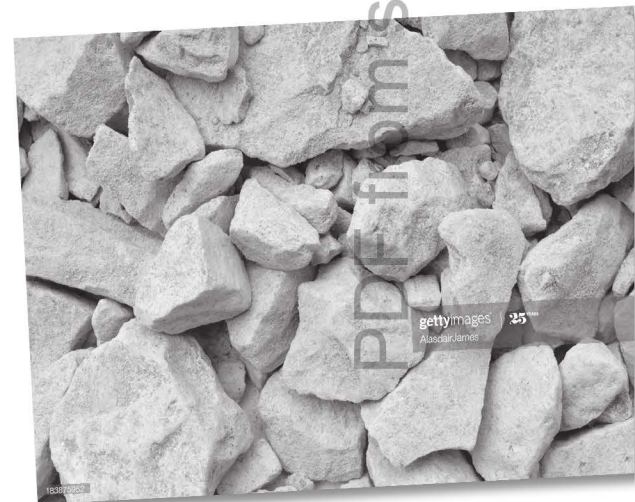
Focus

- 1 Underline or **highlight** the correct alternative in these sentences about sedimentary rocks.
 - a Heat and ice break up rocks in a process called **erosion/ weathering**.
 - b A river **erodes/weathers** the bottom and sides of its valley.
 - c The river **deposits/transport**s sediments on the sea bed.
 - d Layers of sediments build up on the sea floor in a process called **accumulation/sedimentation**.
 - e Over time the **top/bottom** layers of sediments become rock.
 - f Sediments in sandstone are more **fine grained/coarse grained** than in shale.
 - g **Limestone/sandstone** is formed of crushed sea shells.
 - h Fossils are only found in **sandstone/sedimentary** rock.

Practice

- 2 a How do you know that the rock in the picture is sedimentary rock?

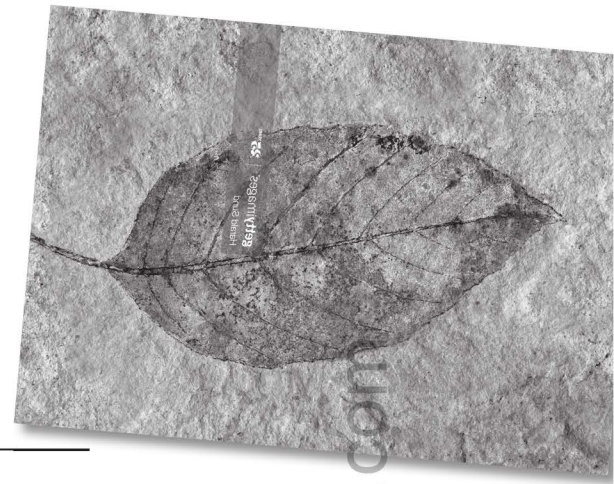
- b How could you decide whether this is limestone or shale?



3 a How did this rock form?

b Is this fossil a mould or a cast?

c Describe how the fossil formed.



Challenge

Read the web page below.

The West Coast Fossil Park

Today there are few plants and animals along the south west coast of South Africa because of the lack of water.

For forty years there was a phosphate mine here. Miners kept finding strange fossilised bones. The museum could not identify the bones as belonging to any animals that live today. Eventually the mine was closed and the area is now a National Heritage site and open air museum.

Scientists found fossils of more than 200 different animals all dating back to five million years ago when a river flowed through the area to the sea. The climate was hot and wet with big trees, ferns and plants.

Scientists have found fossils of huge bears, three-toed horses, different species of elephants and many short-necked giraffe (Sivathere). All these animals are extinct now. They think that some of these animals tried to cross a flooded river and were all drowned. The fast-flowing river washed them downstream where they finally came

41

>

continued...

to rest on some rocks. As their bodies rotted in the sun, hyenas and vultures feasted until only the bones were left behind. When the floodwaters subsided, a waterhole remained and became a popular place for all the animals in the area.

The sea level rose and flooded the valley once again, submerging the rocks and pools in salt water. Another layer of sediments was added. This time the sediments were phosphate-rich sand.

For the next five million years this collection of bones lay undisturbed, making the change from bone to stone.

4 a How old are the fossils in the West Coast Fossil Park?

b How were the bones of the animals 'turned into stone'?

c How do scientists know that the fossilised animals are now extinct?

d How is it possible for scientists to describe the climate, animal and plant life that existed five million years ago in this area?

e What type of sedimentary rock were the fossils found in?

> 3.3 Metamorphic rocks and the rock cycle

Focus

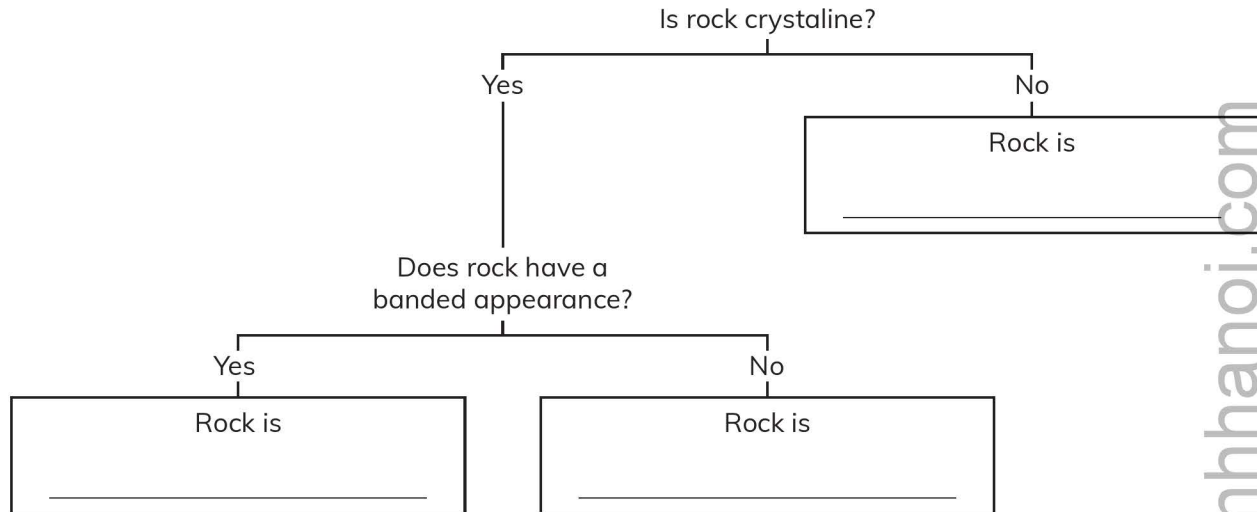
- 1 Circle the correct answer a, b or c.
 - a Metamorphic rocks are formed when:
 - A *magma cools and becomes rock*
 - B *existing rocks are changed by heat and pressure*
 - C *sediments build up on the sea bed.*
 - b Metamorphic rocks are:
 - A *very soft*
 - B *in layers*
 - C *very hard and in layers.*
 - c In the rock cycle, different processes are shown by:
 - A *arrows and labels*
 - B *pictures*
 - C *lines.*
- 2 Draw lines between each rock and the metamorphic rock it changes into.

Limestone	Gneiss
Sandstone	Marble
Shale	Slate
Granite	Quartzite

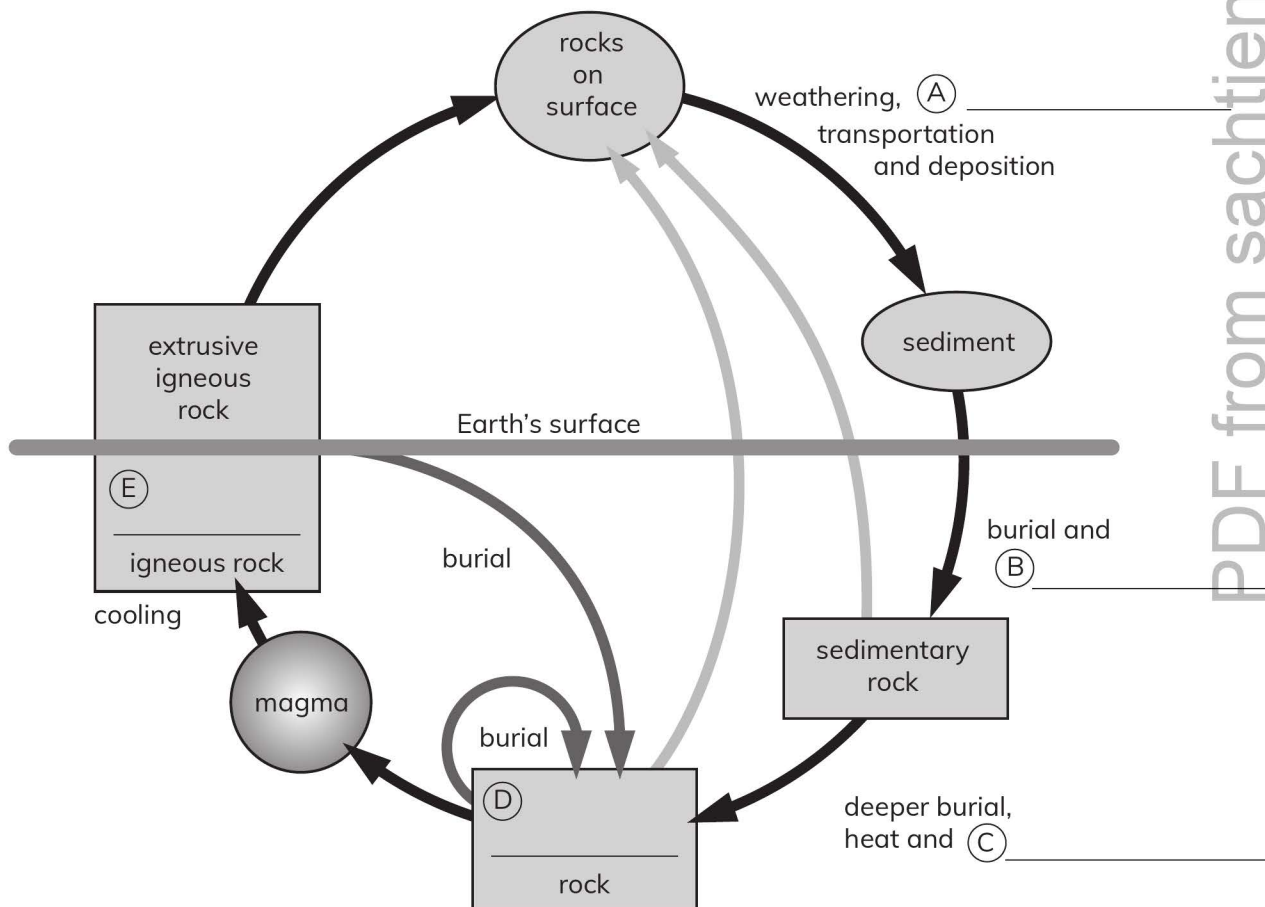
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Practice

- 3 You are given a rock to identify. Use this key to identify the type of rock. Fill in the spaces in the key.



- 4 Fill in the correct words at A–E on this diagram of the rock cycle.



Challenge

- 5 Draw a rock cycle to describe the history of a piece of rock over the last 350 million years. The rock begins and ends as granite.

Include all these words in your drawing:

granite

sedimentary rock

sedimentation

metamorphic rock

heat

pressure

melting

burial

weathering

sandstone

cooling

erosion

transportation

deposition

quartzite

igneous rock

PDF from sachtienganhanoi.com

> 3.4 Soil

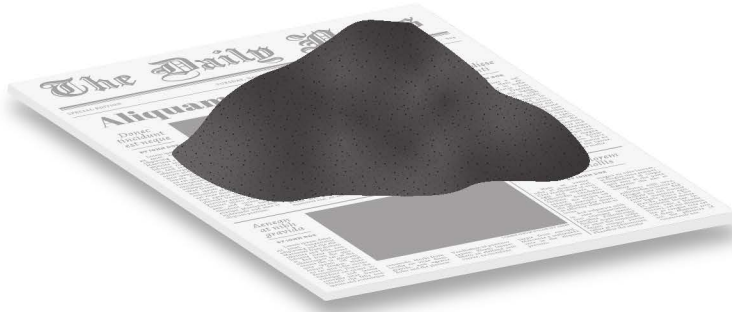
Focus

- 1 Complete these sentences. Use the words from the box.

break soil rocks

The Earth's crust consists of _____. Rain and ice _____
up the rocks. The small rock particles become _____.

2



List four things you can investigate to describe the soil on the newspaper.
We have done the first one as an example.

- a Colour _____
- b _____
- c _____
- d _____

3 List the three main types of soil

a _____

b _____

c _____

Practice

4 Sofia and Zara want to plant vegetables on a small plot of ground.

The soil is hard and dry. The soil is yellowish in colour. There are no bits of dead leaves or animals in it.

a Describe the texture and colour of the soil.

b Does the soil contain organic matter? How do you know this?

c Is this a sandy soil, a clay soil or a loam soil?

d Do you think the soil is suitable for growing vegetables? Explain why or why not.

- e Write down three ways that Sofia and Zara can change the composition of the soil to make it more suitable for growing vegetables.

Challenge

Read the web pages below.



Organic farming

Organic farming uses environmentally friendly farming methods. These methods improve the soil and keep humans healthy because there are no chemicals in the food they produce.

Organic farmers always rotate their crops. This means they do not plant the same crop on the same soil over and over again. They use compost instead of artificial fertilisers. Instead of using pesticides they use biological pest control.

What is biological pest control?

The most natural way to control pests eating plants is to make sure there are natural predators of the pests. These natural predators can be insects such as ladybirds, bees and spiders, or lizards, frogs or birds. Organic farmers plant flowers near the crops and hedges at the sides of the field which attract these natural predators.

For example, aphids are a common pest because they eat leaves and stems and fruit. If the farmer makes sure there are always ladybirds on the farm, the ladybirds will eat the aphids. It is possible to buy ladybirds for this purpose! In the photograph you can see a ladybird eating up all the aphids.



- 5 a Describe two ways organic farmers maintain the composition of their soil.

- b What is biological pest control?

- c How do organic farmers use biological pest control?

- d Find out the names of two birds and two animals that are natural predators of snails.

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IMAGE**

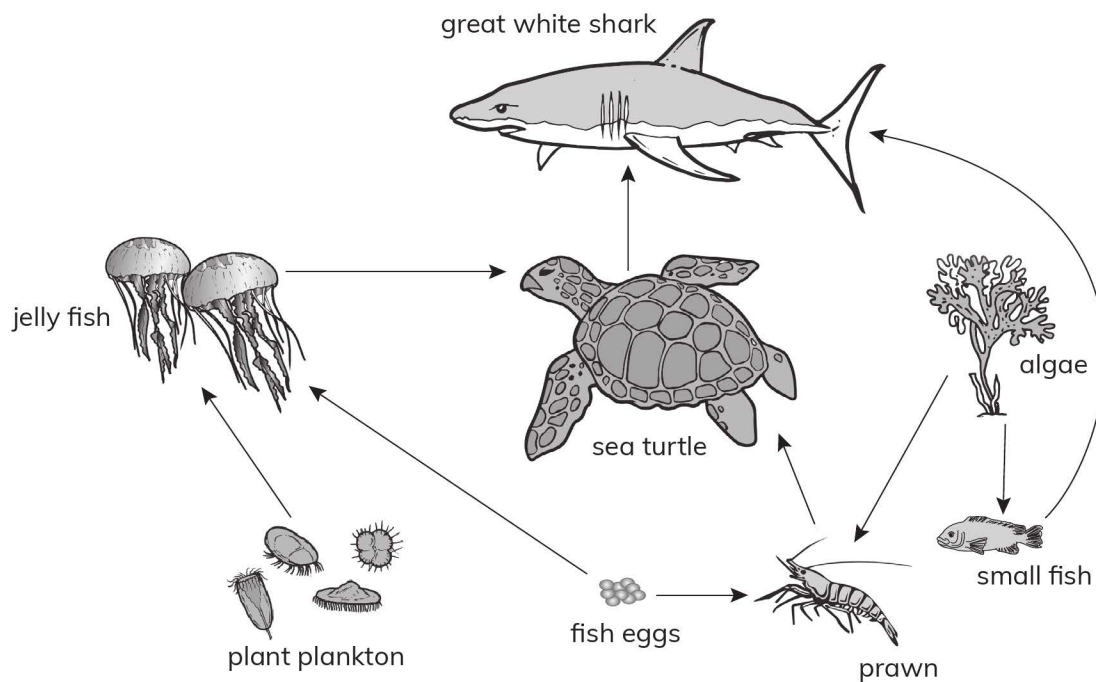
4

Food chains and food webs

> 4.1 Food chains, food webs and energy transfers

Focus

1 Look at the drawing of an ocean food web.



- a Find and circle each of the food chains listed.
Use a different colour for each food chain.

fish eggs → jelly fish → sea turtle

algae → small fish → great white shark

plant plankton → jelly fish → sea turtle → great white shark

- b What do the arrows in the food chains show?

Practice

Look at the drawing of an ocean food web in the Focus exercise.

- 2 a Identify and draw two other food chains with four links.

- b Name the producers in the food web.

- c Identify and name each of the following types of consumers in the food web:

- i herbivores

- ii omnivores

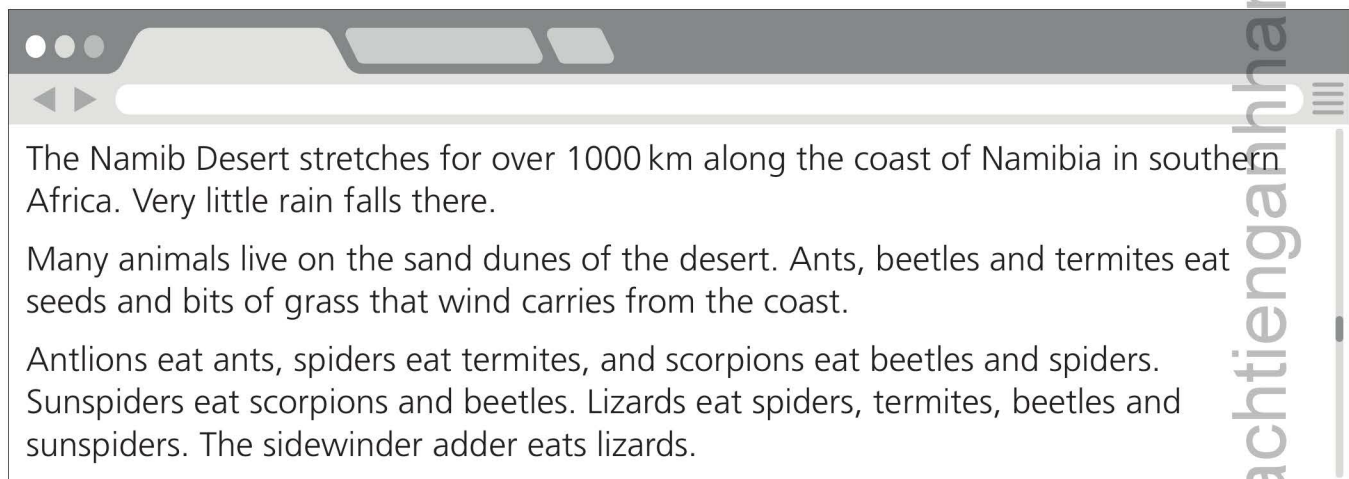
- iii carnivores

- d What is the source of energy for all food chains and food webs?

- e Predict what would happen to the sea turtles if all the great white sharks were killed.

Challenge

Read the web page below.



- 3 a In words, write ten food chains from the information given in the passage above. One of the food chains should have six links.

- 1

- 2

- 3

- 4

- 5

- 6

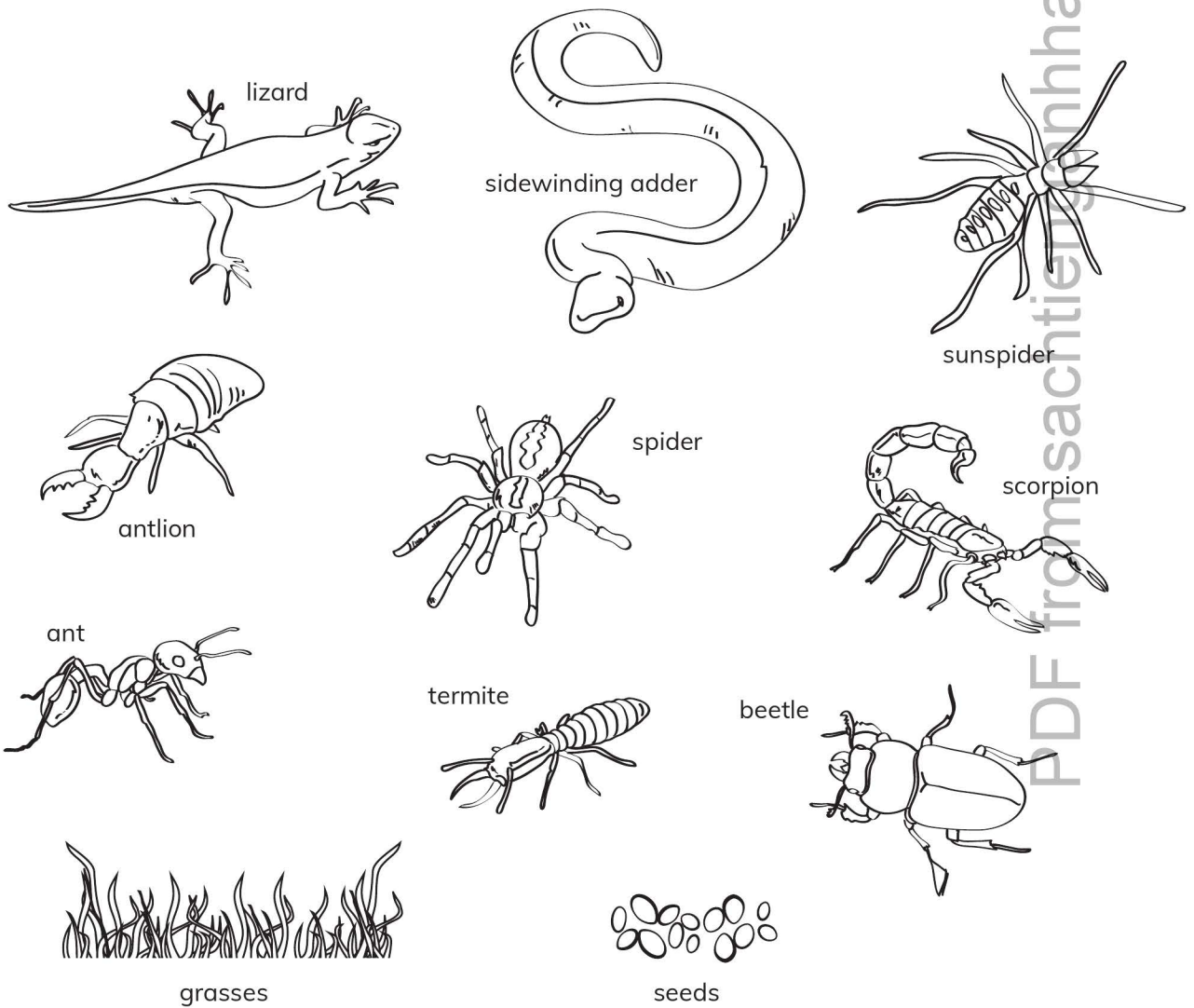
7 _____

6 _____

9 _____

10 _____

- b Draw arrows on the drawing to make a food web from the different food chains you have written.



- c Explain why a food web is more like what happens in nature than a food chain.

> 4.2 Harm to food chains and food webs

Focus

- 1 Find and circle the word in the word search grid that has each of the following meanings. Circle each word you find and write the word next to its meaning:
 - a harmful or poisonous
 - b to build up or increase
 - c chemicals used by farmers to get rid of insects and other living things that eat their crops
 - d the air, land and water around us
 - e a harmful metal in food chains

p	w	e	r	t	y	n	u	m	o	a
e	n	v	i	r	o	n	m	e	n	t
s	d	f	g	h	g	f	t	r	n	l
t	d	g	u	t	o	x	i	c	e	t
i	v	b	n	m	e	l	h	u	j	x
c	f	d	t	w	e	i	u	r	m	j
i	q	a	h	c	v	b	d	y	p	r
d	f	m	j	y	d	d	c	j	a	u
e	a	c	c	u	m	u	l	a	t	e

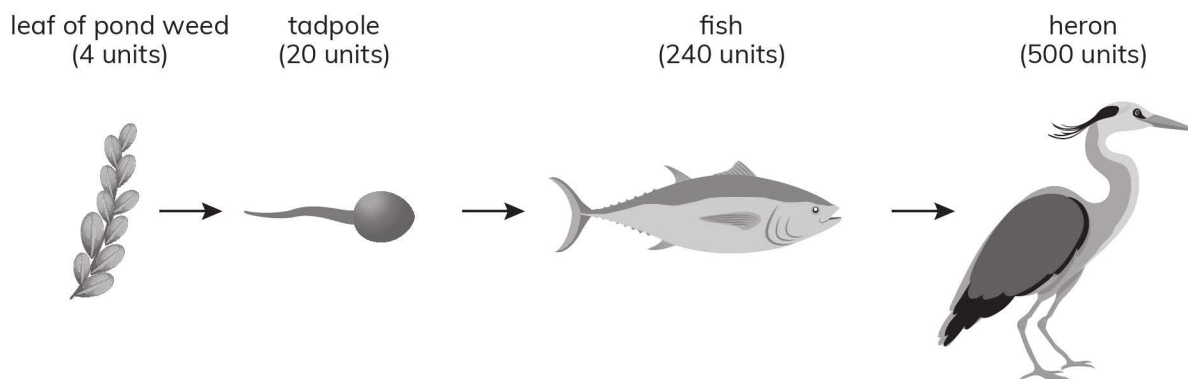
2 Decide if each of the sentences is true or false.

Tick a box (✓) to show your answer.

	True	False
a Harmful substances can damage living things in food chains.	☐	☐
b Harmful substances move through food chains instead of energy.	☐	☐
c Harmful substances in food chains cannot affect humans.	☐	☐
d Some harmful substances in a food chain break down in the bodies of the living things.	☐	☐

Practice

3 A factory has dumped a harmful chemical into a river. The drawing shows a food chain in the river and the number of units of the harmful chemical X in each living thing.



a Suggest a way that chemical X entered the food chain.

b Which living thing contained the most units of chemical X?

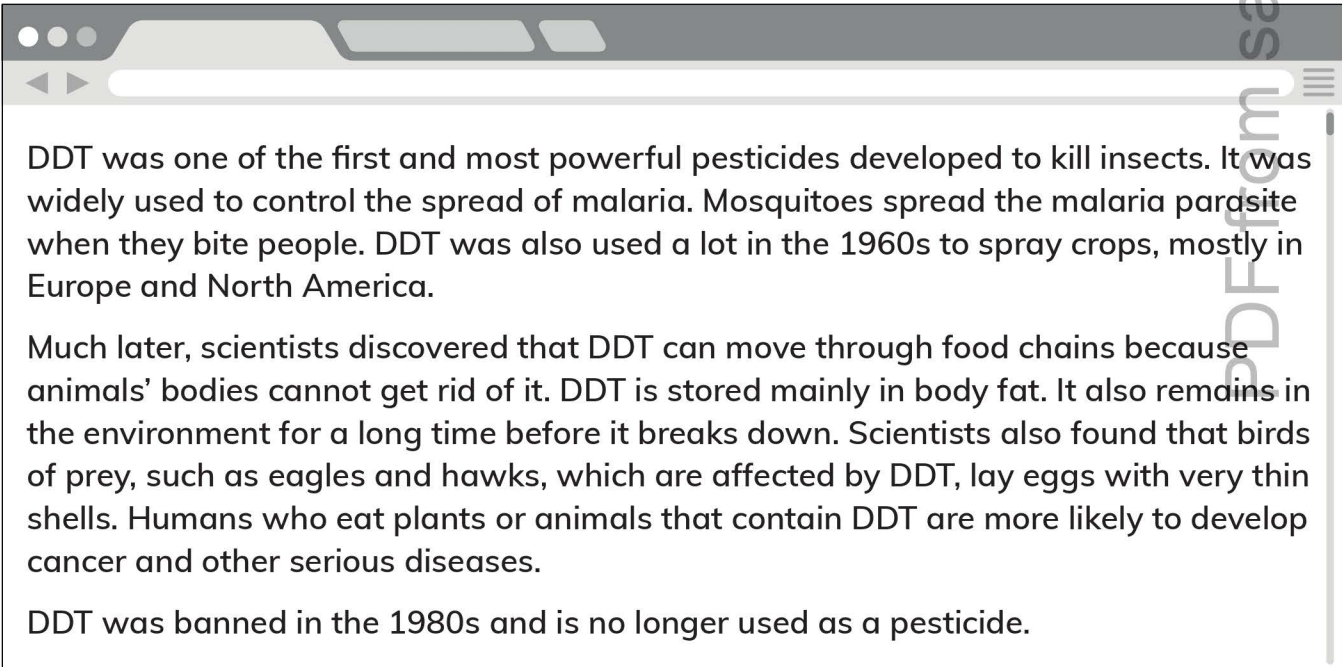
- c How many times more time units are there in the fish than in a pond weed leaf?

- d i What do you notice about the amount of chemical X in a living thing and the position of the living thing in the food chain?

- ii Explain why this is so.

Challenge

Read the web page below.



DDT was one of the first and most powerful pesticides developed to kill insects. It was widely used to control the spread of malaria. Mosquitoes spread the malaria parasite when they bite people. DDT was also used a lot in the 1960s to spray crops, mostly in Europe and North America.

Much later, scientists discovered that DDT can move through food chains because animals' bodies cannot get rid of it. DDT is stored mainly in body fat. It also remains in the environment for a long time before it breaks down. Scientists also found that birds of prey, such as eagles and hawks, which are affected by DDT, lay eggs with very thin shells. Humans who eat plants or animals that contain DDT are more likely to develop cancer and other serious diseases.

DDT was banned in the 1980s and is no longer used as a pesticide.

4 a Why was DDT used a lot in the past?

b i Explain how DDT can enter a food chain.

ii Why can DDT move through food chains?

c i Suggest two reasons why arctic animals such as seals and polar bears often have high levels of DDT in their bodies, even though DDT is no longer used.

ii Some people living in arctic regions eat seals. Why can this be dangerous for them?

- d In a food chain affected by DDT, explain why a frog that eats three locusts does not die, but an owl that eats three frogs dies. What is the word that describes this?

- e Why do you think the number of birds of prey in Europe and North America decreased a lot in the 1960s?

- f Do some research to find out the full name of DDT.

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5

Forces and electricity

> 5.1 Mass and weight

Focus

- 1 a Name the instrument used to measure mass.

- b Name the units mass is measured in.

- c Name the instrument used to measure weight.

- d Name the unit weight is measured in.

- 2 Complete these sentences about mass and weight:

Mass is the amount of _____ in an object.

Weight is the amount of _____ on an object caused by

the force of _____ .

Practice

- 3 Read the following statements about mass and weight.

Mark each statement as true (✓) or false (✗).

- a Mass and weight are the same.
- b We measure mass in kilograms.
- c We measure weight in Newtons.
- d Weight is the amount of matter in an object.

- 4 Class 6 measured the mass and weight of four objects. Complete their table of results.

Remember to include the correct units for each measurement.

Object	Mass	Weight
Brick	1	
Bag of books		25
Bag of potatoes	10	
Bag of cement		400

- 5 If you pull on an object hanging from a forcemeter, would this give you an accurate reading of the object's weight? Say why or why not.

Challenge

6 a What is this instrument called?

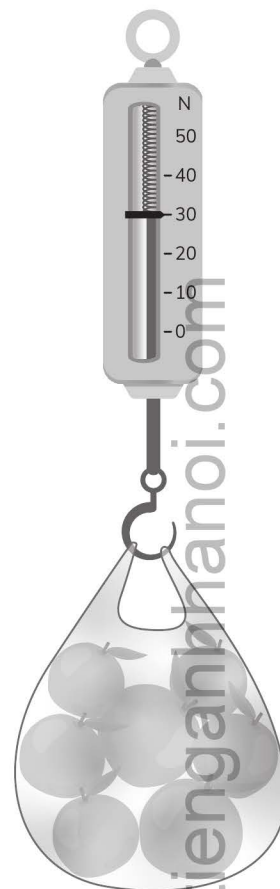
b The instrument is measuring the weight of a bag of oranges.
What does the bag of oranges weigh?

c What is the mass of the bag of oranges?

7 Mr Large has a mass of 90 kg on Earth. He travels to Planet Zogg where the gravity is only half as strong as it is on Earth.

What will Mr Large's weight be on Planet Zogg?

Explain your answer.



> 5.2 The effects of forces

Focus

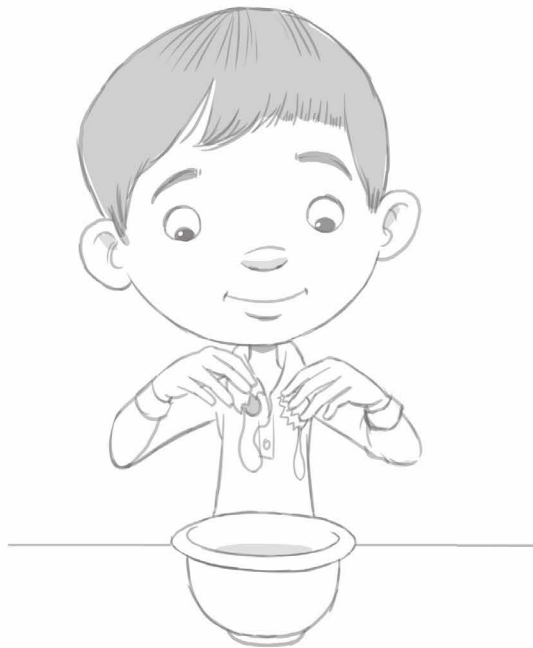
1 In each of the following examples, identify the effect of the force on the object. Choose from:

- Makes an object move.
- Changes the direction of a moving object.
- Changes the shape of an object.
- Changes the speed of an object.

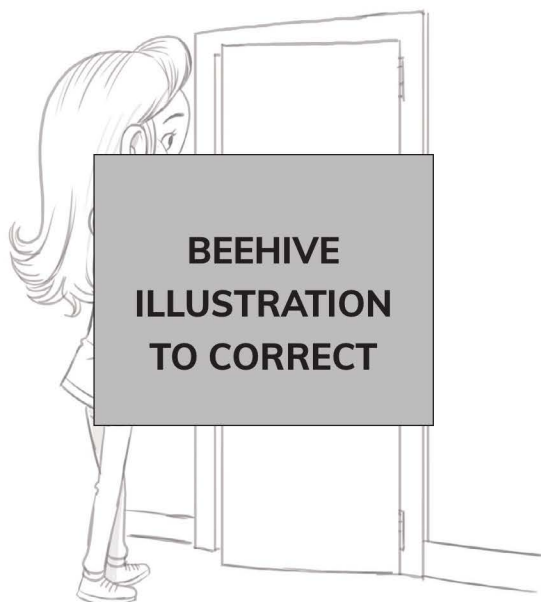
- a Zara pedals her bike faster.
-



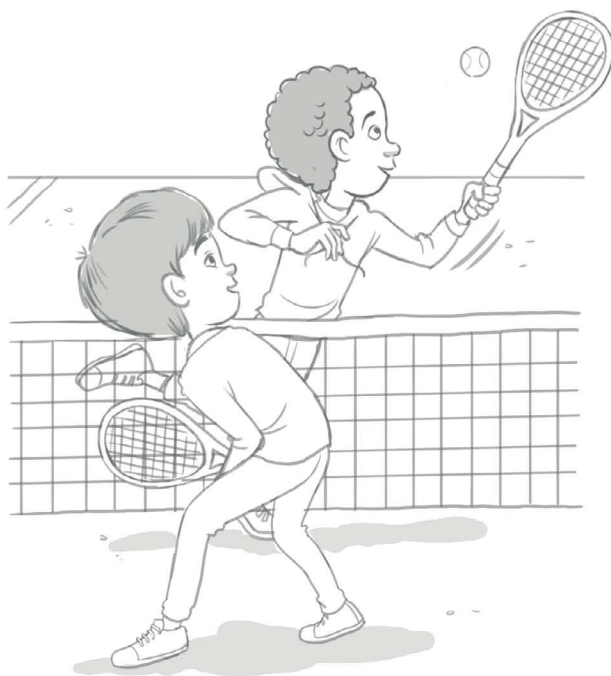
- b Arun cracks an egg into a bowl.
-



- c Sofia pushes open a door.



- d Marcus hits a tennis ball back to Arun.



Practice

2 Write down alongside each picture how the soccer ball is affected by forces applied to it.

a



b



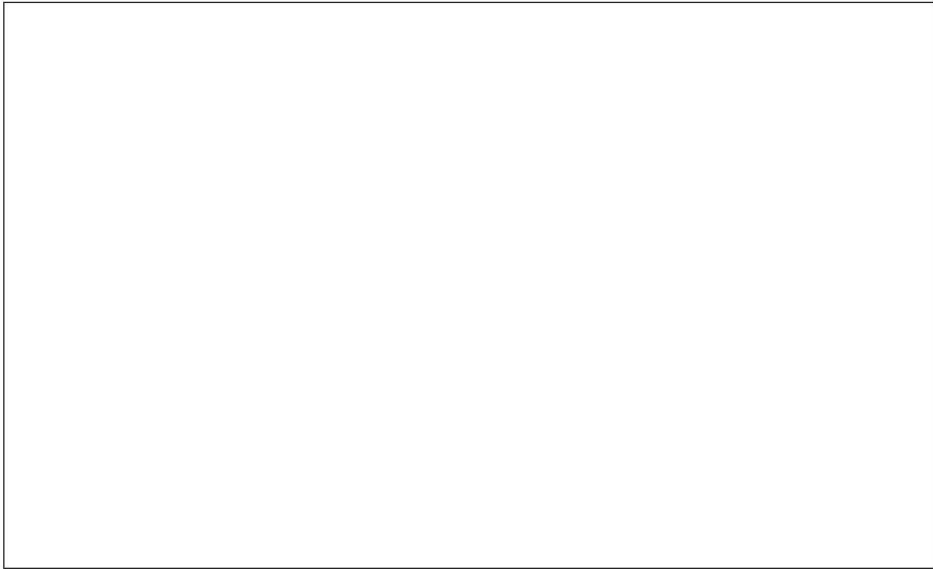
c



d



- 3 Draw a force diagram in the space to show the two forces that affect the soccer ball when it is resting on the ground.



Challenge

- 4 a You have to demonstrate five different effects of forces.

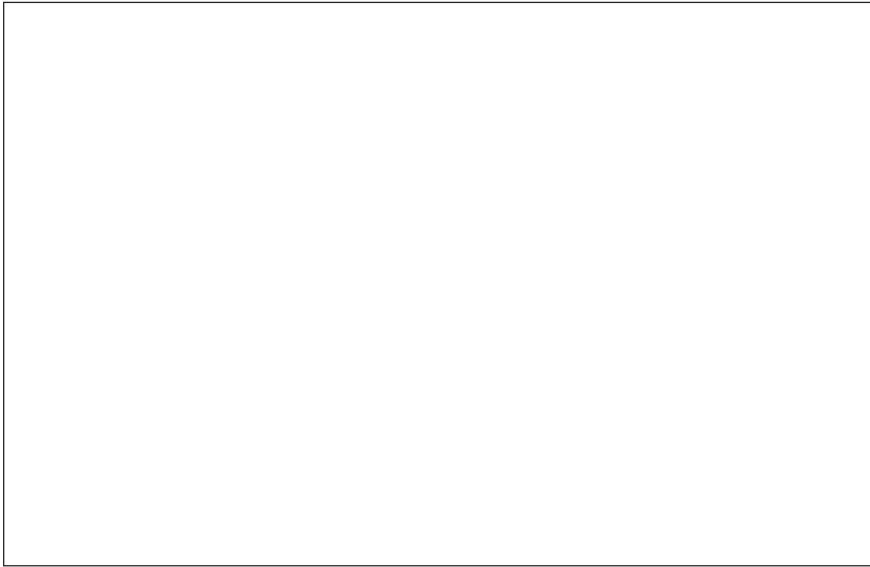
You are given a space on the floor next to a wall, a tennis ball and a strip of sandpaper.

Fill in the actions you would do and the effect of the force on the tennis ball in the table below.

The first one is done as an example.

Action with tennis ball	Effect of force
Roll tennis ball across floor	Make tennis ball move

- b Draw a force diagram to show the forces involved in one of the actions you listed on the table.



> 5.3 Floating and sinking

Focus

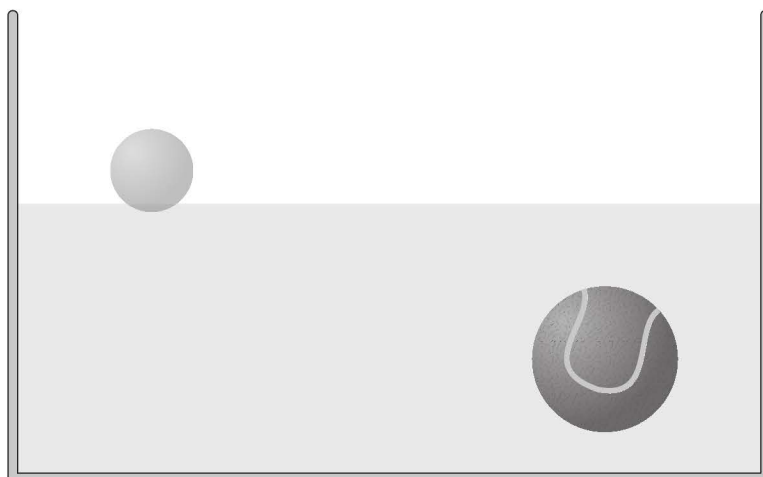
- 1 Complete the sentences by filling in the correct words from the box.

mass displaced float sink upthrust equal less

An object will _____ if the mass of the object is less than, or _____ to, the mass of water displaced by _____.

An object will _____ if the mass of the object is greater than the mass of water _____ by upthrust.

- 2 Complete the diagram with labelled arrows to show why the ping pong ball floats and the tennis ball sinks towards the bottom of the tank.



Practice

Read the web page below.

◀ ▶

Sofia and Zara are doing an investigation to see how mass affects floating and sinking.

They have a glass container of water. They have made a 'raft' out of a square piece of flat polystyrene which floats on the surface. They have a balance.

'Let's try putting coins on the raft and see how many we put on before the raft sinks,' says Zara.

Sofia puts a pile of different coins on the table. 'We have to use the same sized coins – let's sort them out.'

Zara puts a coin on the balance. Each coin has a mass of 5 g.

PDF from Sachtienganhanoi.com

- 3 a What were the control, the independent and the dependent variables in their investigation?

- b Why did the polystyrene 'raft' float on the water?

- c The girls put seven coins on the raft before it sank. Why did the raft sink?

- d Suggest how the girls could have changed the raft and put more coins on it before it sank.

- e Draw a force diagram to show the raft beginning to sink.

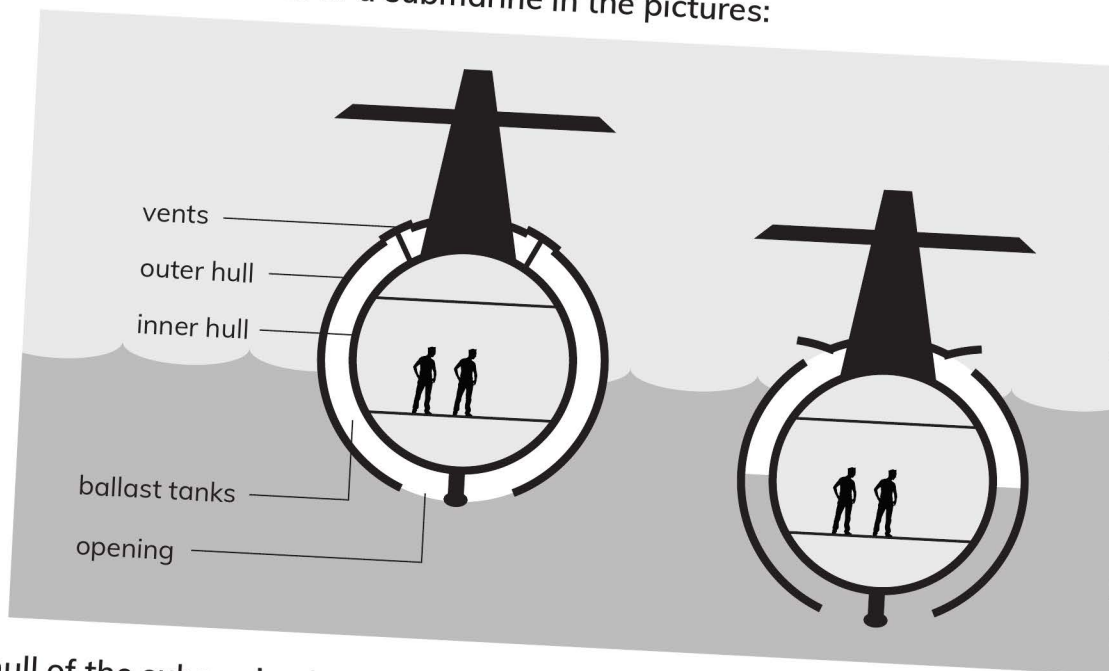


PDF from sachtienganhanoi.com

Challenge

Submarines

Submarines are able to float or sink. How do they manage to do this?
Look at the cut sections of a submarine in the pictures:



The hull of the submarine has two layers. In between the layers there are big storage spaces. These are called the ballast tanks. When the submarine is floating, the ballast tanks are filled with air. This makes the total mass of the submarine less than the mass of the water it displaces, so the submarine floats.

When the submarine needs to sink, it opens the doors of the ballast tanks and lets in water. The water takes the place of the air.



- 4 a Explain how filling the ballast tanks with water causes the submarine to sink.

- b Draw a section of the submarine, like the ones above, but when it is completely submerged. Show the forces involved with arrows.

- 5 Find out the answer to each of the following questions.

- a How can the people inside the submarine breathe when the submarine is submerged?

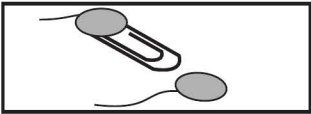
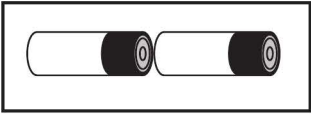
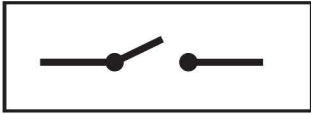
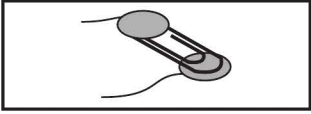
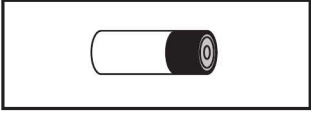
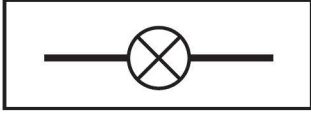
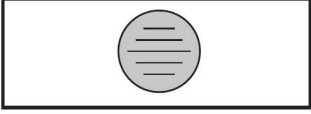
- b How long can a submarine remain submerged?

- c What is the longest time a submarine has remained submerged?

> 5.4 Different circuits and circuit diagrams

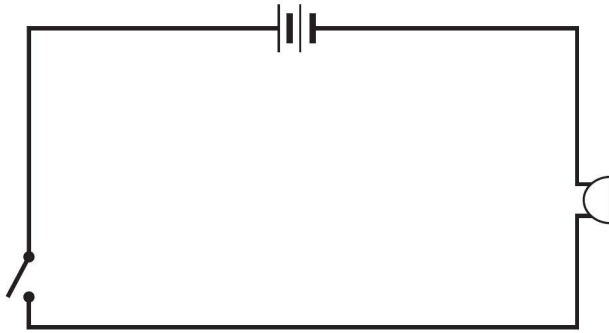
Focus

- 1 Draw lines linking each component with its symbol.

Component	Symbol
	
	
	
	
	
	
	
	

Practice

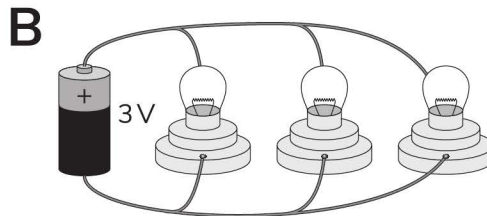
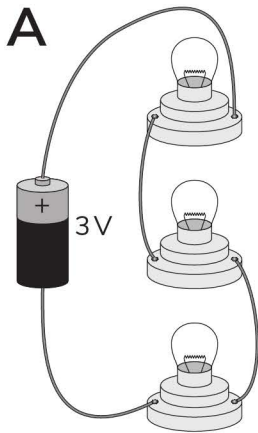
2



- a List the components in this circuit.

- b Will the buzzer come on? Explain why or why not.

- 3 Look at the two circuits, A and B.

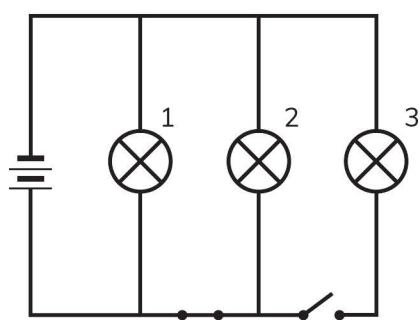


- a Which circuit is a series circuit and which circuit is a parallel circuit?

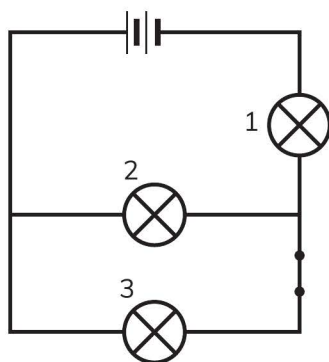
- b In which circuit, A or B, will the lamps shine more brightly? Explain why.

- c Draw circuit diagrams for circuits A and B.

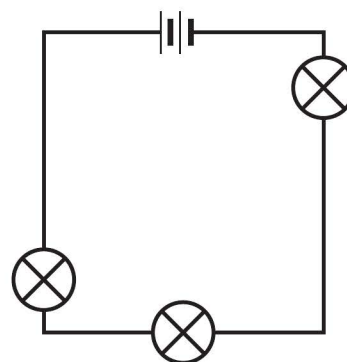
Challenge



A



B



C

- 4 a Which circuits are parallel circuits and which are series circuits?

- b In which circuit will the lamps shine more brightly – circuit B or C? Explain your answer.

- c In circuit B, which lamp will shine the brightest – 1, 2 or 3? Explain your answer.

- d In circuit A, which lamp or lamps will not light up? Explain your answer.

- e In circuit A, what would happen if you opened both the switches?

- f Draw a parallel circuit with a 3V battery, two switches and two buzzers.

6

Light and the Solar System

> 6.1 Reflection

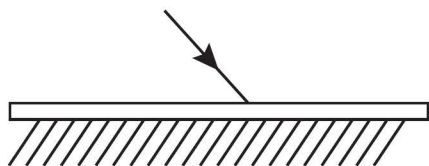
Focus

- 1 The diagrams A–D show light reflecting off plane mirrors. Complete each of the diagrams by adding the missing ray.

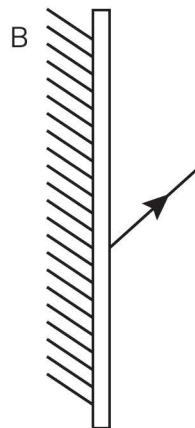
Measure angles with a protractor.

Label all the rays as the incident ray or the reflected ray.

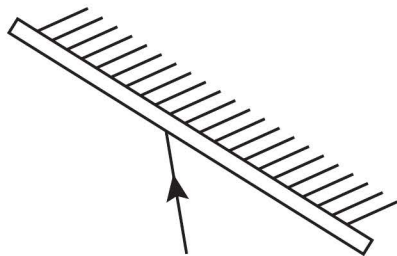
A



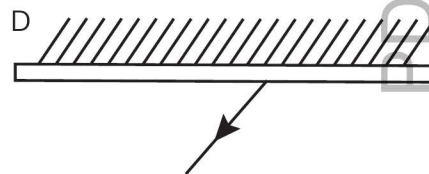
B



C

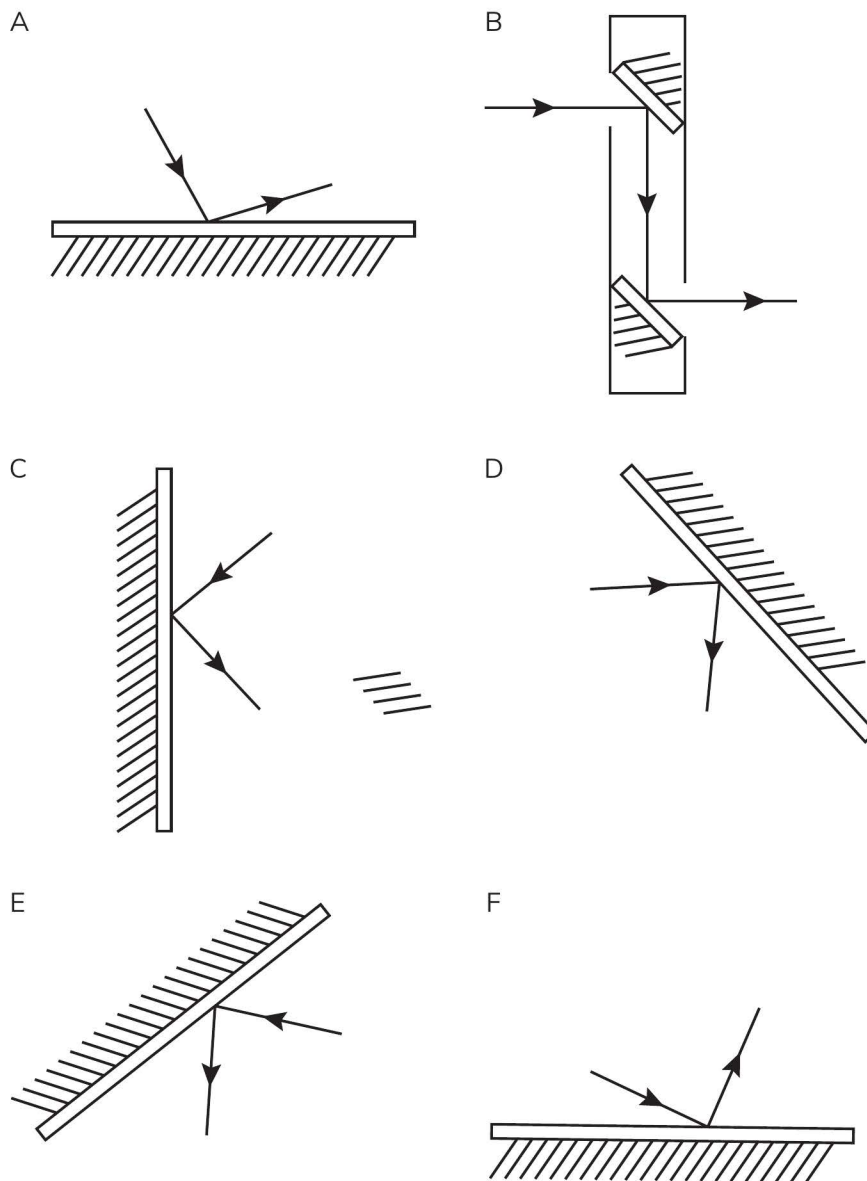


D



Practice

2 Diagrams A–F show light reflecting off plane mirrors.



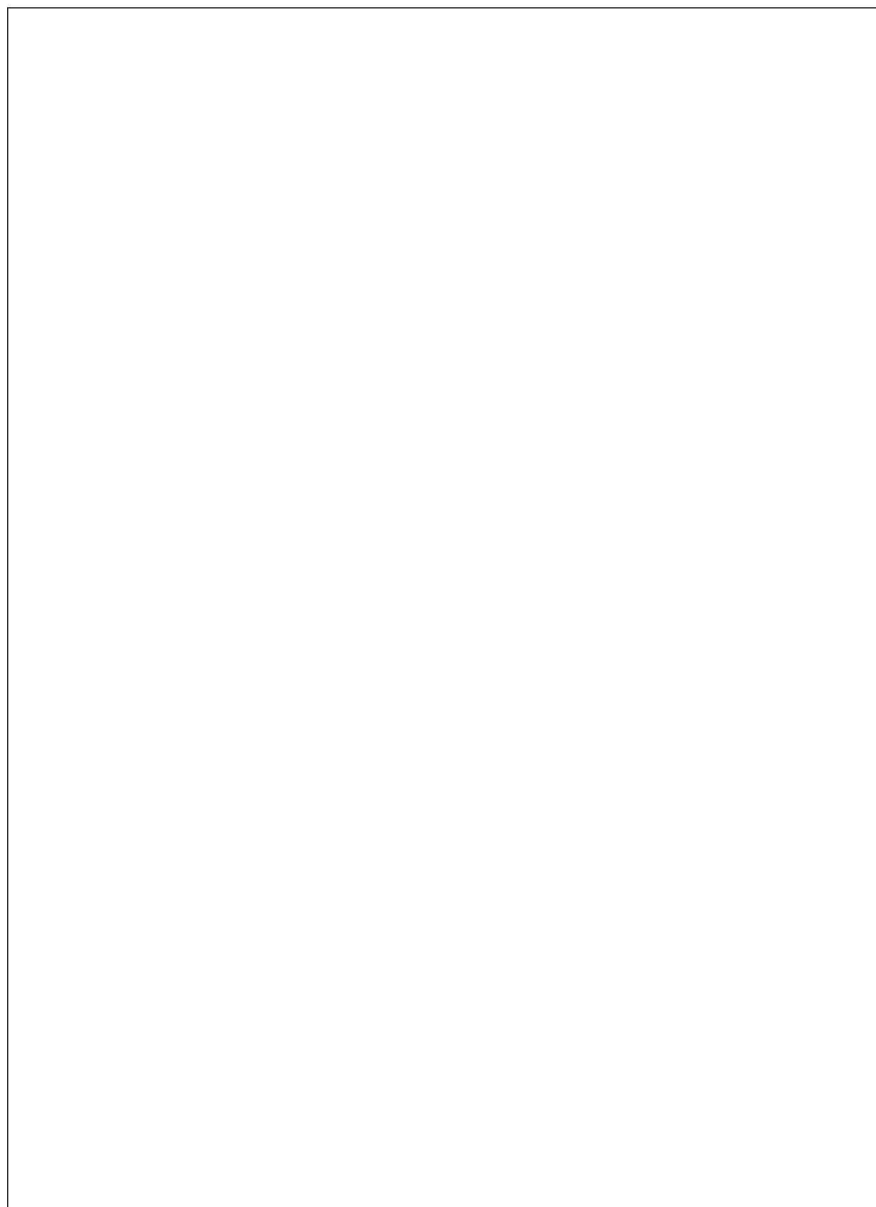
a Which diagram shows a periscope?

b Which two diagrams are incorrect?

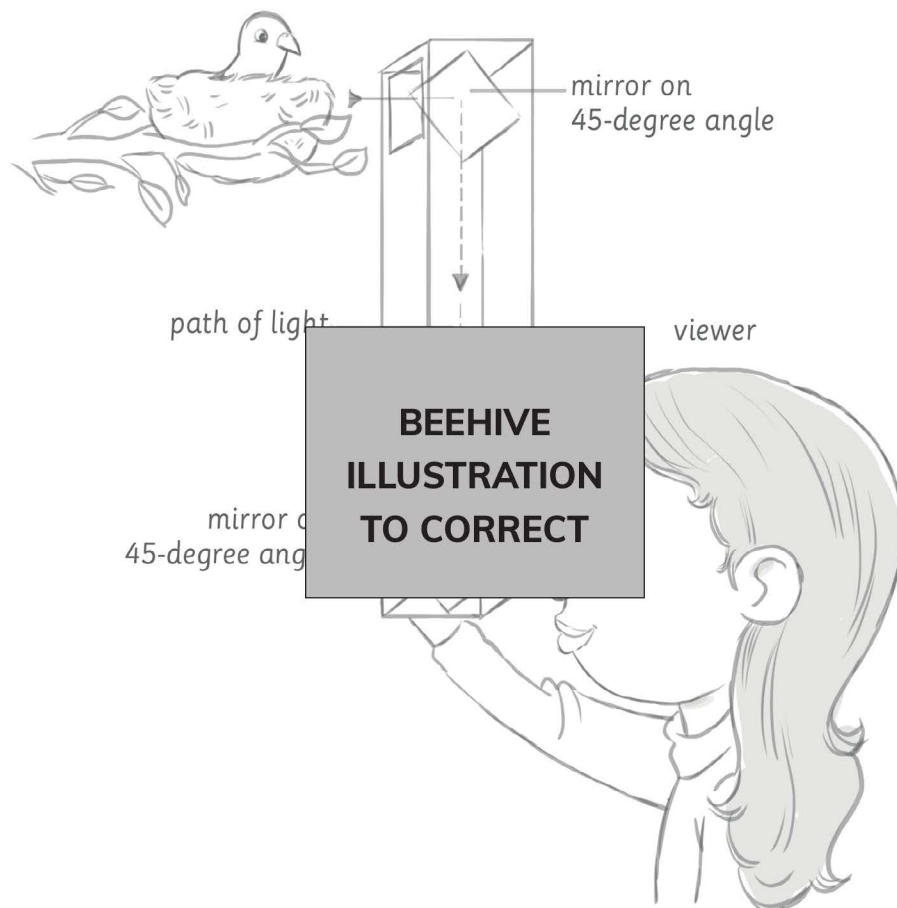
- c Re-draw the two diagrams that are incorrect in the space below.

Measure your angles with a protractor.

Label the incident ray and the reflected ray on each diagram.



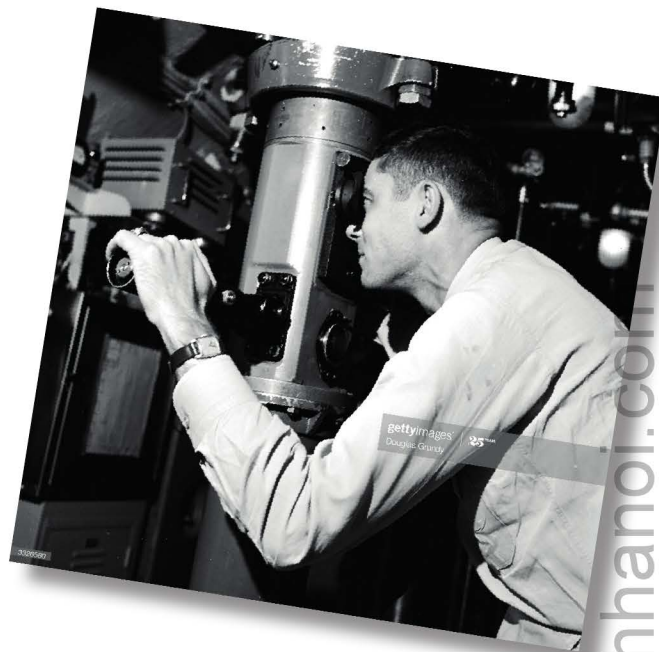
Challenge



- 3 a What is Zara using the periscope for?

- b Describe how this periscope works.

- 4 a The man in the photograph is in a submerged submarine. He is using a periscope.



What do you think he is using the periscope for?

- b Suggest other ways in which periscopes are useful.

> 6.2 Refraction

Focus

- 1 Explain why the pencil in the picture appears to be bent. Cross out the incorrect alternatives in the sentences below.

The pencil is bent because of reflection/refraction.

A ray of light passes from the pencil through the water/air to the glass. The ray bends/straightens when it passes through the glass to the air/water and into our eyes.

We see the bent pencil as a trick/optical illusion.



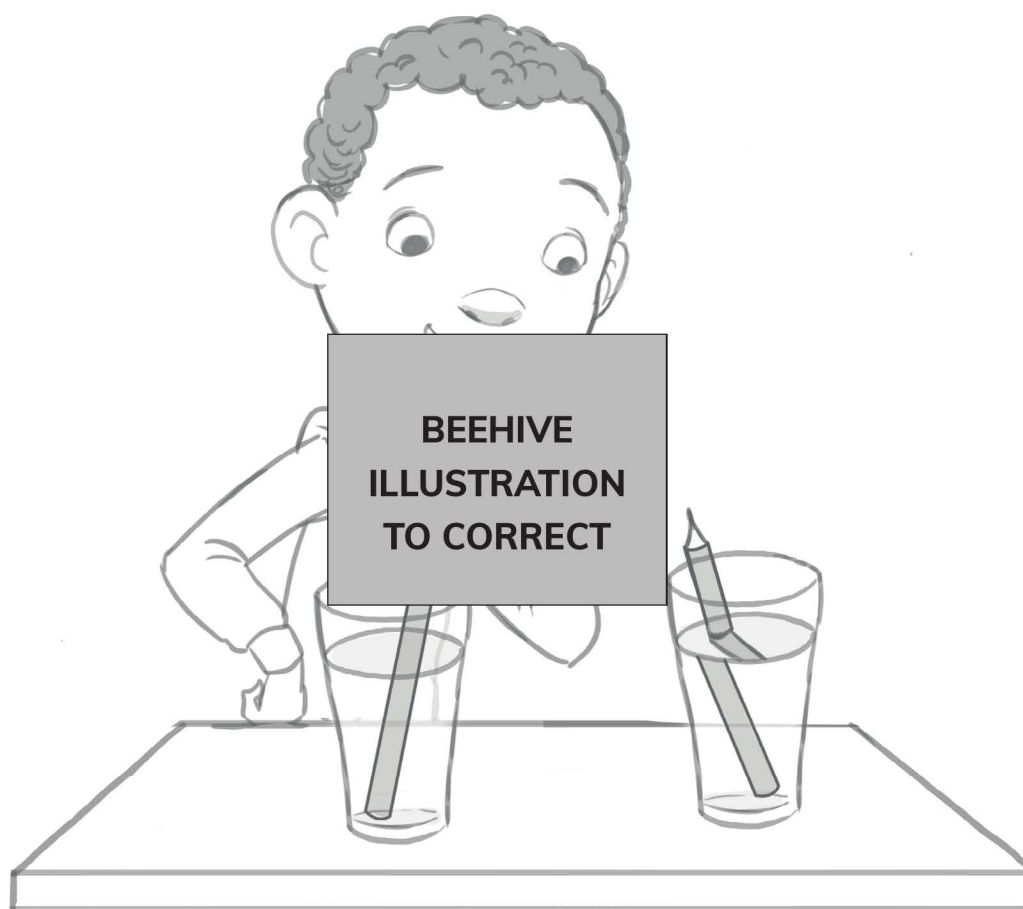
- 2 What happens when you shine a ray of light through a prism?
-

- 3 Cross out the incorrect alternatives in the sentences about lenses below.

A lens is a transparent piece of glass or plastic with at least one straight/curved surface.

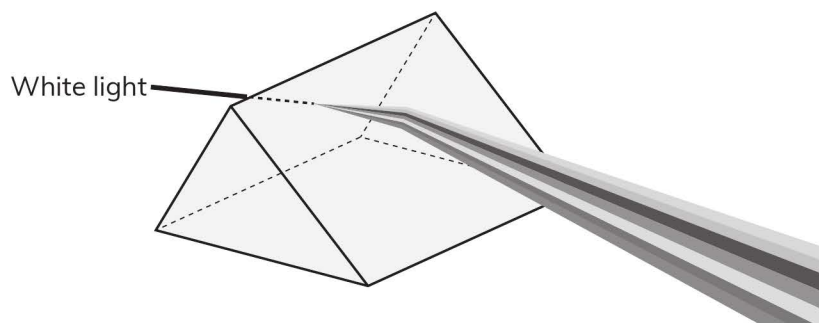
A convex lens makes things look bigger/smaller because light rays bend inwards/outwards as they leave the lens.

Practice



- 4 Draw in the light ray on the diagram above to show how light travels from the pencil to Marcus. Draw a dotted line to show the optical illusion that Marcus sees.

5 The diagram below shows white light splitting into colours.



a What is the triangular glass object?

b How does it affect light passing through it?

c Label the colours on the diagram.

6 a What type of lenses are used in binoculars?

b Explain why.

Challenge

Read the web page below.

A screenshot of a web browser window. The address bar is empty. The main content area contains the following text:

Have you ever read a story or seen a movie where someone is trying to walk across a desert and is desperate for water? They keep seeing water in the distance only to find it is just more sand.

Look at the photograph taken in the Namib desert. It really looks as though the thorn trees are reflected in a lake of water ... but it is just sand.

This is a mirage and it is an optical illusion.

gettyimages®

Pete Turner

25 YEARS

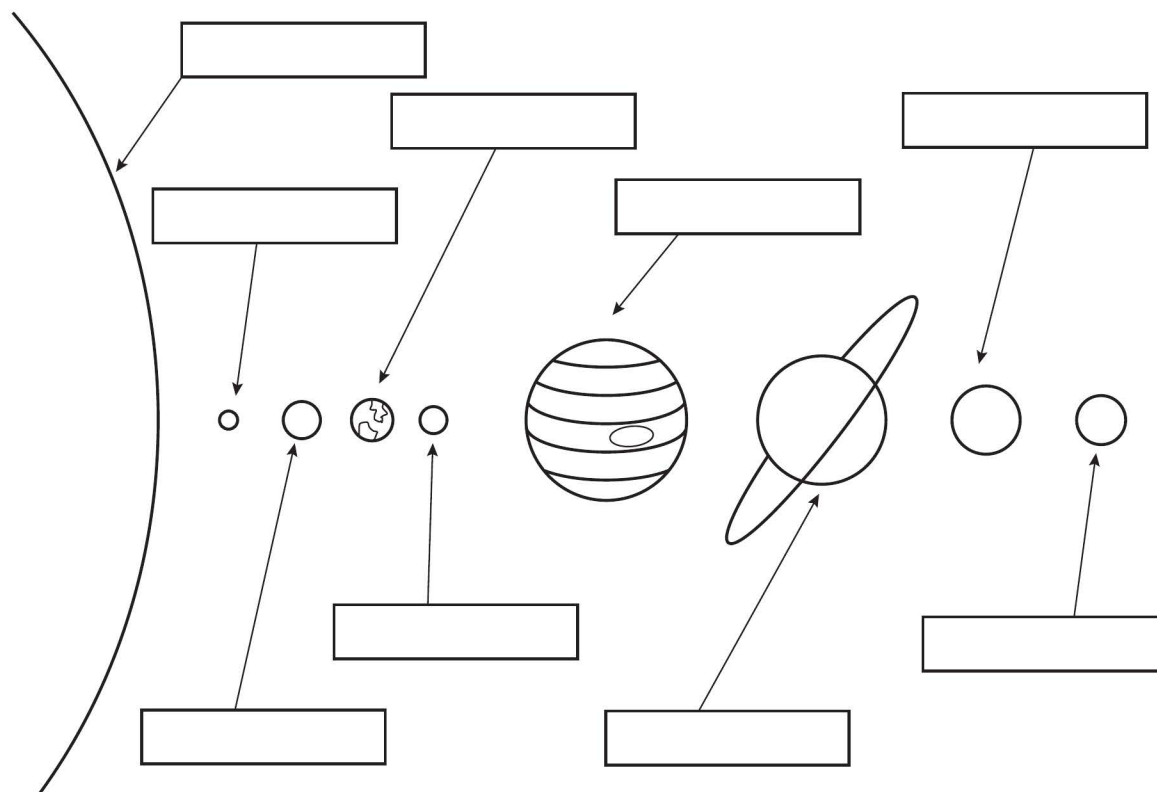
PDF from sachtienganhanoi.com

- 7 Use the internet or reference books to find out how a mirage happens.

> 6.3 The Solar System

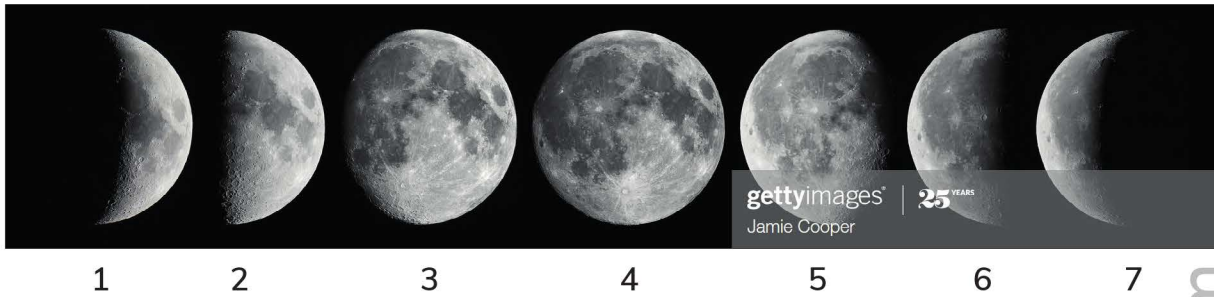
Focus

- 1 a Label the Sun and the eight planets in the boxes on the diagram.



- b Name the two movements that all the planets make.

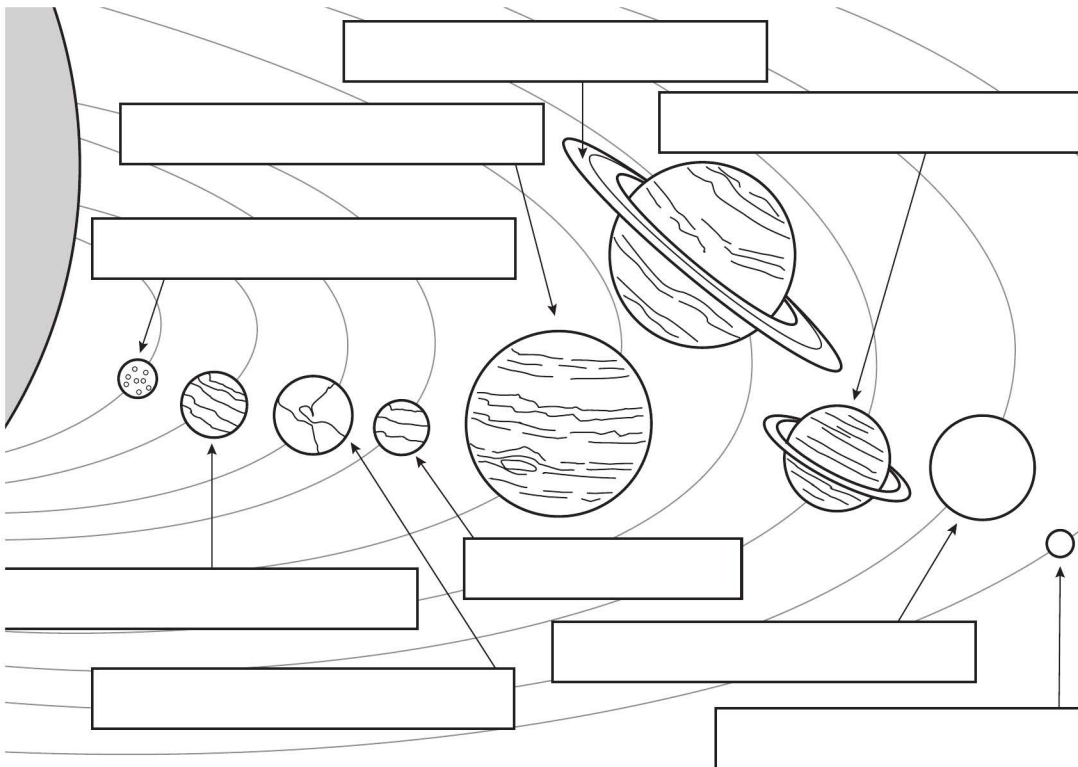
2



- a What movement causes the Moon to have different shapes at different times?
- _____
- b How long does it take for the Moon to make this movement?
- _____
- c Between which numbers is the Moon waning?
- _____

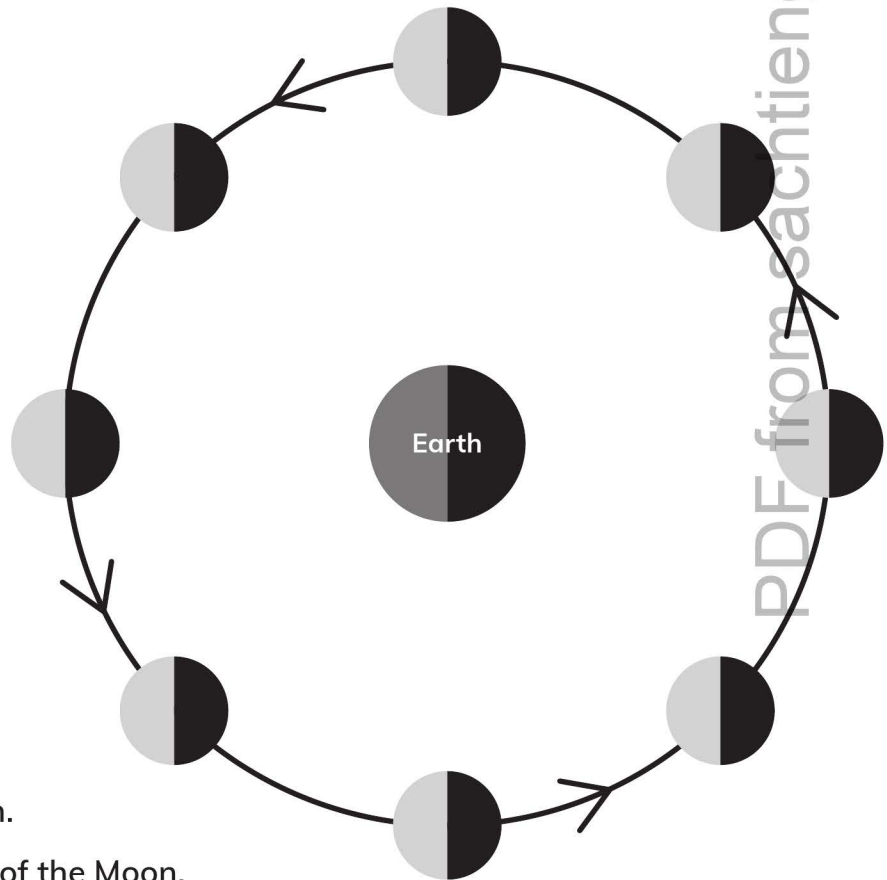
Practice

3



- a Name the planets on the diagram
- b How long are an 'earth day' and an 'earth year'?
- c Which planets take more than an earth year to complete their orbit around the Sun? Explain why.
- d Which planet has the shortest year? Explain why.
- e Why does this diagram not show the solar system accurately?

4



On the diagram:

- a Draw and label the Sun.
- b Label the eight phases of the Moon.

Challenge

Read the web page below.

Compare the planets Earth, Mars and Jupiter

Mars is 240 million km from the Sun compared to Earth, which is 150 million km from the Sun. Mars is often called 'the red planet' – the red colour is due to iron in the rocks. Mars' atmosphere consists mainly of carbon dioxide. In winter temperatures fall to -125°C . On parts of Earth temperatures drop to -50°C .

Mars has two moons, called Phobos and Deimos.

Mars takes 25 Earth hours to make a complete turn on its axis and 687 Earth days to complete an orbit around the Sun.

Spacecraft that have landed on the surface of Mars reported a rocky surface with frequent dust storms and no life.

Jupiter is composed almost entirely of gases such as hydrogen and helium – it is sometimes called a 'gas giant'. The temperature is about -153°C . Jupiter is 800 million km away from the Sun. It takes 12 Earth years to orbit the Sun.

Jupiter takes 10 Earth hours to make a complete turn on its axis.

Jupiter has 16 known moons. Four of these moons are larger than the rest and easily visible to us on Earth through telescopes. One of Jupiter's moons, called Io, is about the same size as our Moon. Ganymede is the largest moon in the Solar System and is in fact bigger than Mercury.

Juno spacecraft is orbiting Jupiter until 2021 and sending amazing photographs of swirling gases back to Earth.



- 5 a Use the information about Mars and Jupiter to complete this table:

Planet	Earth	Mars	Jupiter
Distance from Sun			
Time taken to make one turn on axis			
Time taken to make one orbit round the Sun			
Number of moons			
Lowest temperature in winter			
What is planet made of?			

- b Identify a pattern in the time taken to make one orbit round the Sun for the three planets. Explain the pattern.

- c Give two reasons why it is possible for plants, animals and humans to live on Earth.

- d Give two reasons why it is not possible for plants, animals and humans to live on Jupiter.

- e Suppose it was possible for people from Earth to live on Mars. What would be familiar about daily life?

- 6 The four biggest moons of Jupiter are often called the 'Galilean Moons' after the astronomer Galileo Galilei. Find out more about Galileo and answer these questions.

- a Where and when was Galileo born?

- b How did Galileo manage to find out more about the planets?

- c What else, besides the four biggest moons of Jupiter, did Galileo identify for the first time in the Solar System?

- d Why was Galileo put under house arrest?
